Introduction to Operating Systems

4. Command Line Interface

2026-10-28

USO Team

Bibliography

1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 2 - *Basic Commands and Directory Hierarchy*
     - Sections 2.1 - 2.2 - *The Bourne Shell*, *Using the Shell*
     - Section 2.4.2 - *Shell Globbing ("Wildcards")*
     - Sections 2.8 - 2.10 - *Environment and Shell Variables*, *The Command Path*, *Special Characters*
     - Section 2.16 - *Listing and Manipulating Processes*
   - Chapter 3 - *Devices*
     - Section 3.4 - *Device Name Summary* (terminals)
   - Chapter 11 - *Introduction to Shell Scripts*
     - Section 11.2 - *Quoting and Literals*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 4 - *Procese*
     - Section 4.3.3 - *Foreground și background*
     - Sections 4.5.2, 4.6.1
   - Chapter 7 - *Interfața în linia de comandă*
     - Sections 7.1, 7.2.3 - 7.2.5
   - Chapter 9 - *Pornirea sistemului*
     - Section 9.6.1 - *Pornirea terminalelor de login*

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It was a mistake to think that GUIs ever would, could, or even should, eliminate CLIs.

Jeffrey Snover (architect of Windows PowerShell)

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Command Line Interface

- Terminal & Shell
- Foreground & Background
- Sessions and Terminals
- Signals
- Environment Variables
- The Command Line
- Quoting, Escaping and Globbing
- Running a Command

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Terminal & Shell

How we talk to the operating system

Bibliography

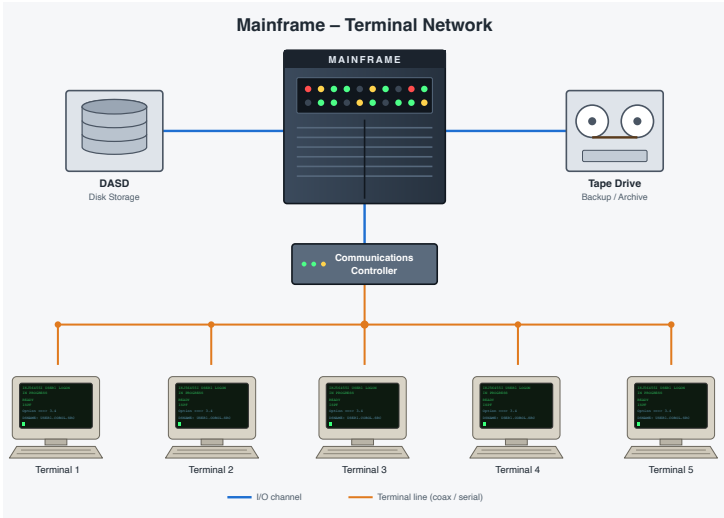
For this section

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The *Classic* Terminal

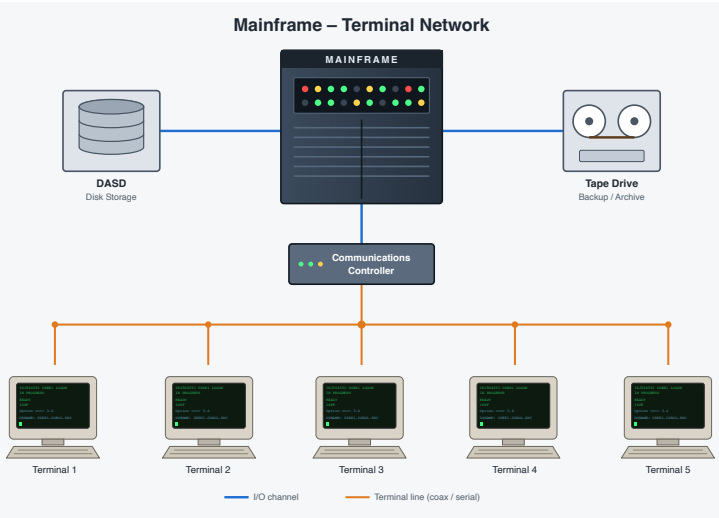
Many users, one computer



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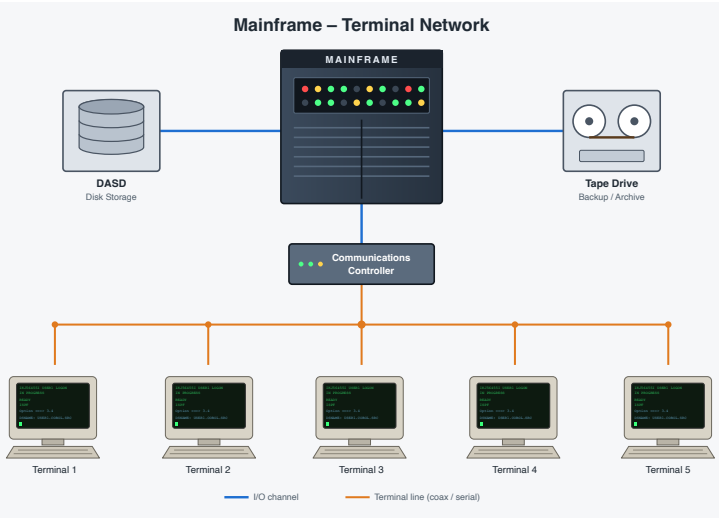


- one large computer (the *mainframe*), many users

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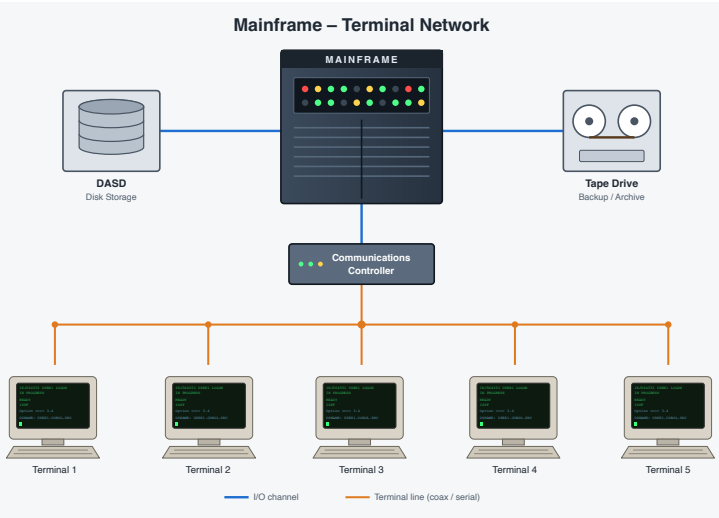


- one large computer (the *mainframe*), many users
- every user has a **terminal**: a keyboard and a printer or a screen, **no processor**

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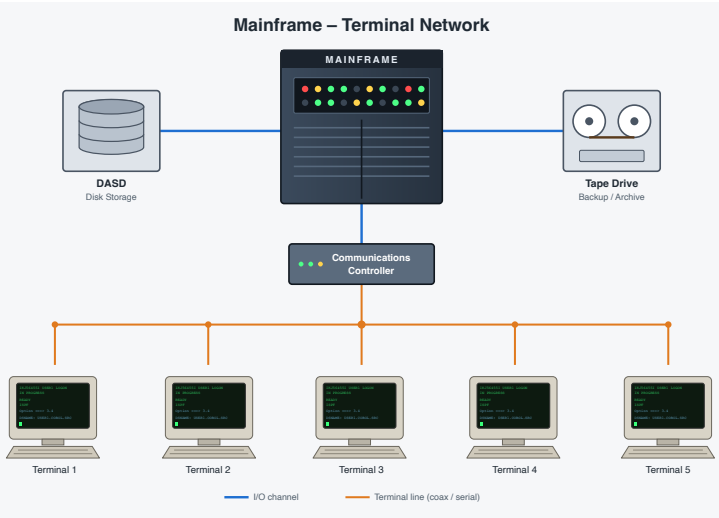


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- the terminals were *TeleTypes* (**TTY**): they send the keys to the computer and print what it sends back

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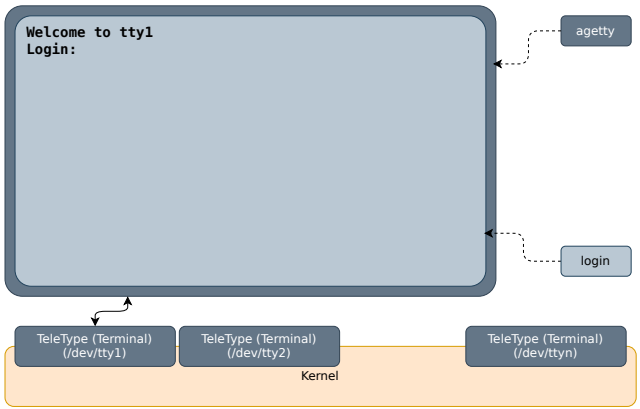
Many users, one computer



- one large computer (the *mainframe*), many users
- every user has a **terminal**: a keyboard and a printer or a screen, **no processor**
- the terminals were *TeleTypes* (**TTY**): they send the keys to the computer and print what it sends back
- 💡 Linux still calls terminals `tty`: `/dev/tty1`, `/dev/tty2`, ...

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The *Real* Terminal

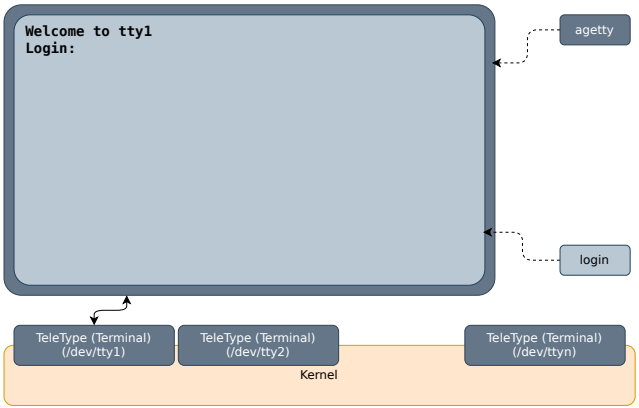



```
[PID 1]  systemd (init)
     # The first user-space process
     # started by the kernel
```

- the kernel provides several *virtual terminals*: `/dev/tty1` ... `/dev/tty6`
- they share the same screen and keyboard

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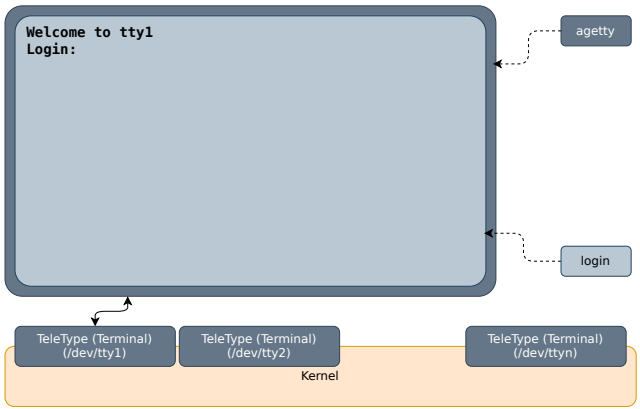



```
[PID 1]  systemd (init)
    # The first user-space process
    # started by the kernel
└── [PID 20+]  agetty tty1
        # Waits for a login on /dev/tty1
```

- the kernel provides several *virtual terminals*: `/dev/tty1` ... `/dev/tty6`
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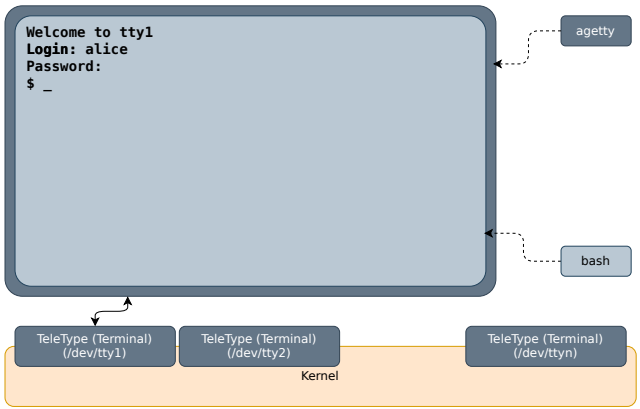



```
[PID 1]  systemd (init)
     # The first user-space process
     # started by the kernel
├── [PID 20+]  agetty tty1
├── [PID 21+]  agetty tty2
├── [PID ...]  ...
└── [PID 25+]  agetty tty6
       # One for every text terminal
```

- the kernel provides several *virtual terminals*: `/dev/tty1` ... `/dev/tty6`
- they share the same screen and keyboard
- `Alt` + `F1` ... `F6` changes the terminal

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The *Real* Terminal



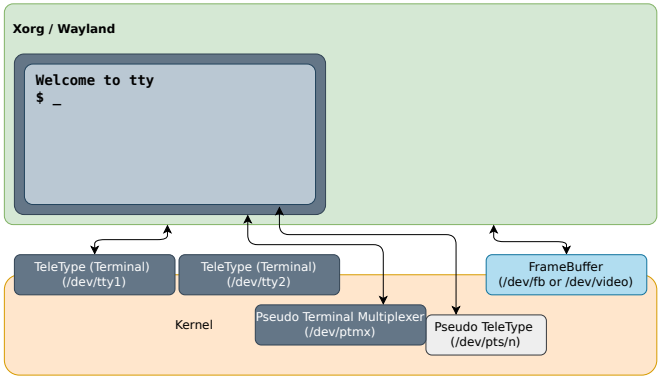

```
[PID 1]  systemd (init)
     # The first user-space process
     # started by the kernel
├── [PID 20+]  login alice
│      # agetty became login (exec)
│      └── [PID 30+]  bash
│             # alice's shell, on /dev/tty1
├── [PID 21+]  agetty tty2
├── [PID ...]  ...
└── [PID 25+]  agetty tty6
```

User `alice` logs in using `/dev/tty1`

- the kernel provides several *virtual terminals*: `/dev/tty1` ... `/dev/tty6`
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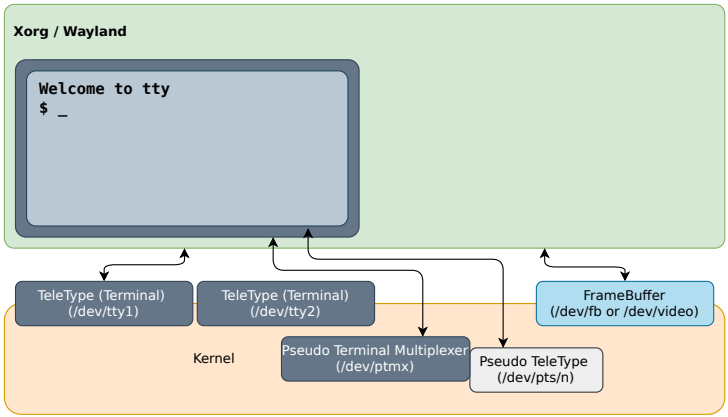
```
[PID 1]  systemd (init)
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│     └── [PID 30+]  bash
├── [PID 21+]  agetty tty2
├── [PID ...]  ...
├── [PID 25+]  agetty tty6
└── [PID 8+]  gdm / sddm / lightdm
       # Graphical login
```

The graphical UI is another application, using one of the terminals

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The *Virtual* Terminal

Graphical Mode Linux



Switch Modes

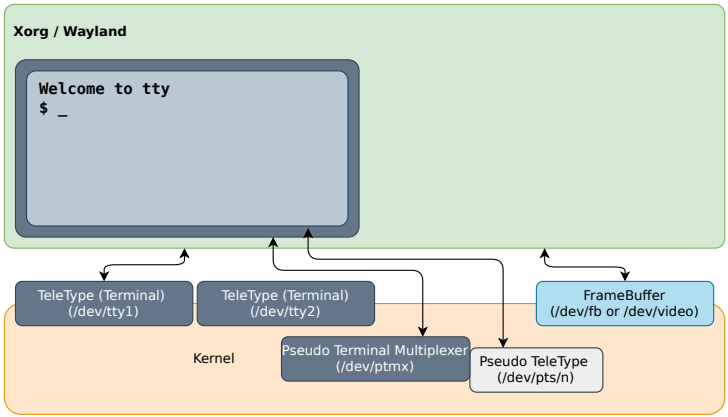
- Ctrl + Alt + F3 ... F6 → text mode
- Ctrl + Alt + F1 or F2 → back to the graphical mode (F7 on older systems)

Xorg or *Wayland* (display servers) take over a terminal (`/dev/tty1` or `/dev/tty2`) and draw on the screen through the framebuffer (`/dev/fb0`)

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The *Virtual* Terminal

Graphical Mode Linux



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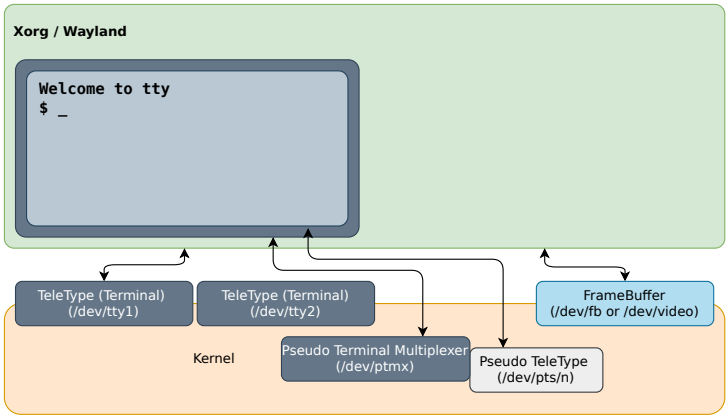
Console Applications

- run in a *terminal emulator* (`kitty`, `ptyxis`, `alacritty`, ...)

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The *Virtual* Terminal

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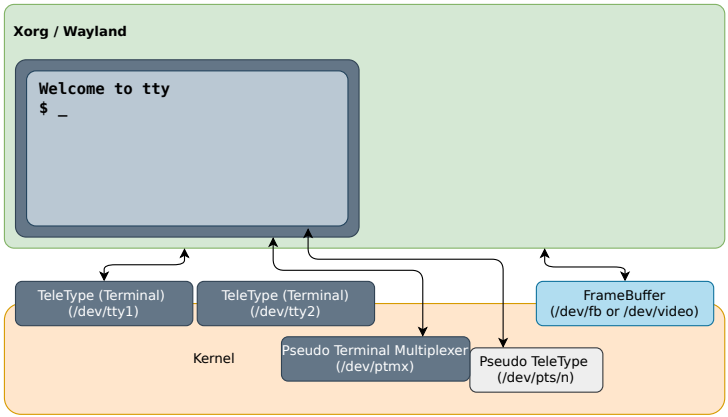
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- run in a *terminal emulator* (`kitty`, `ptyxis`, `alacritty`, ...)
- the terminal emulator is a **graphical** application

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The *Virtual* Terminal

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Console Applications

- run in a *terminal emulator* (`kitty`, `ptyxis`, `alacritty`, ...)
- the terminal emulator is a **graphical** application
- it provides a *pseudo terminal*, `/dev/pts/n`

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Shell

Operating System Interaction



A process whose main purpose is to allow the interaction with the Operating System

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Shell

Operating System Interaction



A process whose main purpose is to allow the interaction with the Operating System

⌨ Command Line

- `bash`, `sh`, `zsh`, `fish`
- `cmd`
- `powershell`

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Shell

Operating System Interaction



A process whose main purpose is to allow the interaction with the Operating System

⌨️ Command Line

- `bash`, `sh`, `zsh`, `fish`
- `cmd`
- `powershell`

🔤 Text User Interface

- `mc`
- `nnn`
- `yazi`

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Shell

Operating System Interaction



A process whose main purpose is to allow the interaction with the Operating System

⌨️ Command Line

- `bash`, `sh`, `zsh`, `fish`
- `cmd`
- `powershell`

🔤 Text User Interface

- `mc`
- `nnn`
- `yazi`

🖱️ Graphical

- `explorer.exe`, Finder, `nautilus`
- Total Commander
- Control Panel, Settings

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Terminal & Shell

Two different programs



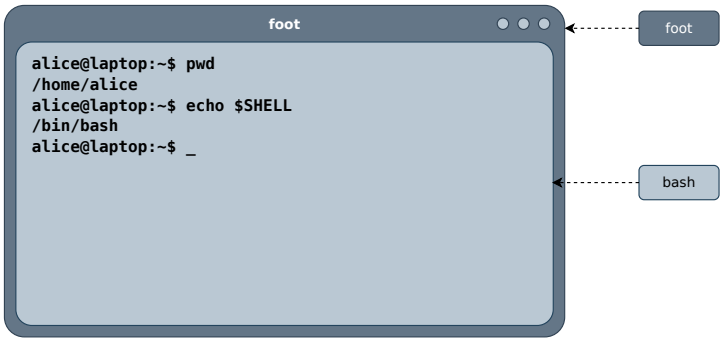
Terminal (`foot`)

- the window, the font, the colors
- reads the keyboard and draws the text

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Terminal & Shell

Two different programs



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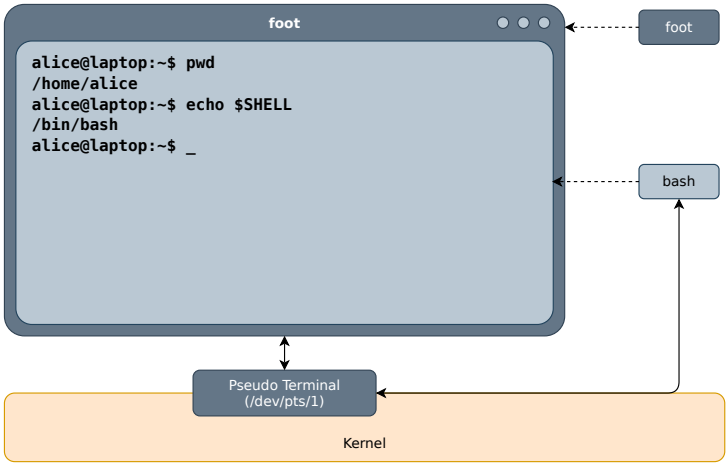
Shell (`bash`)

- the text inside the window: the prompt and the output
- runs the commands

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Terminal & Shell

Two different programs



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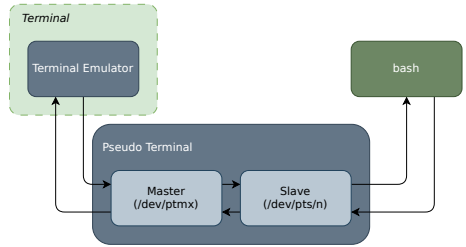
- the text inside the window: the prompt and the output
- runs the commands
- they talk through the *pseudo terminal* `/dev/pts/1`

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Terminal Emulator

Provides a pseudo terminal

`kitty`, `ptyxis`, `alacritty`, ...



`bash`, `sh`, `zsh`, ...

- only see a terminal: `/dev/pts/n`

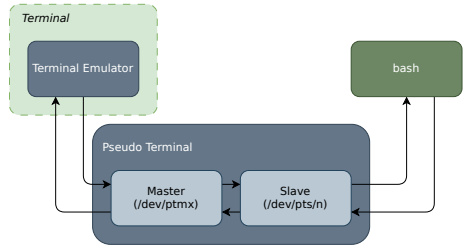
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1. opens `/dev/ptmx`, the kernel creates a *pseudo terminal*: a **master** (kept by the emulator) and a **slave** (`/dev/pts/n`)



`bash`, `sh`, `zsh`, ...

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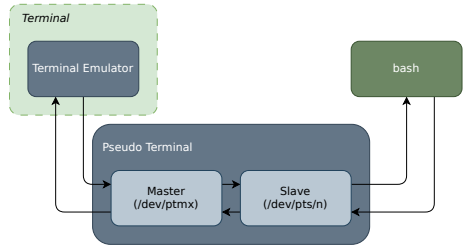
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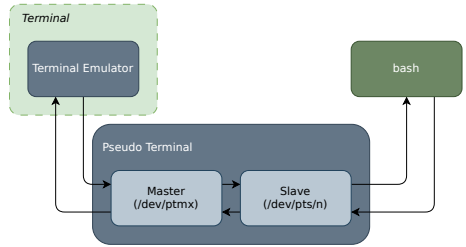
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3. the keys you press are written into the **master**, the shell reads them from `stdin`



`bash`, `sh`, `zsh`, ...

- only see a terminal: `/dev/pts/n`

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Terminal Emulator

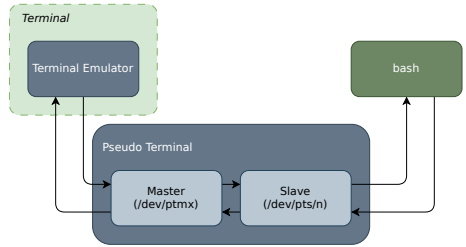
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3. the keys you press are written into the **master**, the shell reads them from `stdin`
4. the shell writes into `stdout`, the emulator reads it from the **master** and draws the text in its window

`bash`, `sh`, `zsh`, ...

- only see a terminal: `/dev/pts/n`



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Terminal Emulator

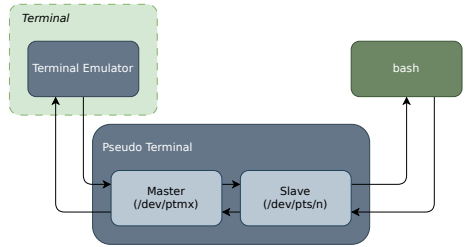
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`bash`, `sh`, `zsh`, ...

- only see a terminal: `/dev/pts/n`




```
$ tty                    # the terminal of the shell
/dev/pts/1
$ ps                     # the processes of this
terminal
    PID TTY          TIME CMD
   4011 pts/1    00:00:00 bash
   4242 pts/1    00:00:00 ps
```

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Signals

notifications for the processes

Bibliography

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     - Section 4.5.2 - *Semnale*

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Signals

✉ the OS sends a message to a process

- a message sent from the OS to a process
- a number
- may have an *extra payload number*

List of signals

```
$ kill -l # sends a signal, does not kill a process
```

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Signals

✉ the OS sends a message to a process

- a message sent from the OS to a process
- a number
- may have an *extra payload number*

List of signals

```
$ kill -l # sends a signal, does not kill a process
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL
 5) SIGTRAP      6) SIGABRT      7) SIGBUS       8) SIGFPE
 9) SIGKILL     10) SIGUSR1     11) SIGSEGV     12) SIGUSR2
13) SIGPIPE     14) SIGALRM     15) SIGTERM     16) SIGSTKFLT
17) SIGCHLD     18) SIGCONT     19) SIGSTOP     20) SIGTSTP
21) SIGTTIN     22) SIGTTOU     23) SIGURG      24) SIGXCPU
25) SIGXFSZ     26) SIGVTALRM   27) SIGPROF     28) SIGWINCH
29) SIGIO       30) SIGPWR      31) SIGSYS      34) SIGRTMIN
```

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Signals

✉ the OS sends a message to a process

- a message sent from the OS to a process
- a number
- may have an *extra payload number*

List of signals

```
$ kill -l # sends a signal, does not kill a process
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL
 5) SIGTRAP      6) SIGABRT      7) SIGBUS       8) SIGFPE
 9) SIGKILL     10) SIGUSR1     11) SIGSEGV     12) SIGUSR2
13) SIGPIPE     14) SIGALRM     15) SIGTERM     16) SIGSTKFLT
17) SIGCHLD     18) SIGCONT     19) SIGSTOP     20) SIGTSTP
21) SIGTTIN     22) SIGTTOU     23) SIGURG      24) SIGXCPU
25) SIGXFSZ     26) SIGVTALRM   27) SIGPROF     28) SIGWINCH
29) SIGIO       30) SIGPWR      31) SIGSYS      34) SIGRTMIN
35) SIGRTMIN+1  36) SIGRTMIN+2  37) SIGRTMIN+3  38) SIGRTMIN+4
39) SIGRTMIN+5  40) SIGRTMIN+6  41) SIGRTMIN+7  42) SIGRTMIN+8
43) SIGRTMIN+9  44) SIGRTMIN+10 45) SIGRTMIN+11 46) SIGRTMIN+12
...             64) SIGRTMAX
```

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Handling Signals

override the default signal handler

When a ✉️ signal arrives, the process will

```
#!/bin/bash

# the signal handler
function no_interruptions_please () {
    echo "No interruptions please"
}

# register the signal handler
trap no_interruptions_please SIGINT

while true; do
  echo "working"
  sleep 1
done

# exit (default action for SIGINT)
```

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Handling Signals

override the default signal handler

When a ✉️ signal arrives, the process will

1. 🛑 stop whatever it is doing
2. 🚥 execute
   - the **default action** (*if no signal handler is registered*)

```
#!/bin/bash

# the signal handler
function no_interruptions_please () {
    echo "No interruptions please"
}

# register the signal handler
trap no_interruptions_please SIGINT

while true; do
  echo "working"
  sleep 1
done

# exit (default action for SIGINT)
```

💡 `Ctrl` + `C` sends `SIGINT`, the default action stops the script

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Handling Signals

override the default signal handler

When a ✉️ signal arrives, the process will

1. 🛑 stop whatever it is doing
2. 🚥 execute
   - the **default action** (*if no signal handler is registered*)
   - OR the **signal handler** and continue its task

```bash
#!/bin/bash

# the signal handler
function no_interruptions_please () {
    echo "No interruptions please"
}

# register the signal handler
trap no_interruptions_please SIGINT

while true; do
  echo "working"
  sleep 1
done

# exit (default action for SIGINT)
```

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Handling Signals

override the default signal handler

When a ✉️ signal arrives, the process will

1. 🛑 stop whatever it is doing
2. 📼 execute
   - the **default action** (*if no signal handler is registered*)
   - OR the **signal handler** and continue its task

```
trap function_name SIGNAL
```

```
#include <signal.h>
void handler(int signum) { /* ... */ }
signal(SIGINT, handler);
```

```
#!/bin/bash

# the signal handler
function no_interruptions_please () {
    echo "No interruptions please"
}

# register the signal handler
trap no_interruptions_please SIGINT

while true; do
  echo "working"
  sleep 1
done

# exit (default action for SIGINT)
```

💡 Ctrl + C now prints the message, the script keeps working

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Default Signal Actions

Default Action	Description
Term	Immediately ends the process. Can usually be caught or ignored.
Term (Unblockable)	Ends the process and **cannot** be caught, blocked, or ignored (`SIGKILL`).
Core	Terminates the process **and creates a core dump** for debugging.
Stop	Suspends the process until it receives `SIGCONT`.
Stop (Unblockable)	Suspends the process, and **cannot** be caught or ignored (`SIGSTOP`).
Cont	Resumes a process that was previously stopped.
Ign	Signal is discarded; no effect on the process.

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Linux Signals: Default Actions (1/2)

#	Signal	Default Action	Description
1	`SIGHUP`	Term	Hangup detected on controlling terminal
2	`SIGINT`	Term	Interrupt from keyboard (`Ctrl` + `C`)
3	`SIGQUIT`	Core	Quit from keyboard (`Ctrl` + `\`)
4	`SIGILL`	Core	Illegal instruction
5	`SIGTRAP`	Core	Trace/breakpoint trap
6	`SIGABRT`	Core	Abort signal from `abort()`
7	`SIGBUS`	Core	Bus error (bad memory access)
8	`SIGFPE`	Core	Floating-point exception
9	`SIGKILL`	Term (Unblockable)	Kill signal
10	`SIGUSR1`	Term	User-defined signal 1
11	`SIGSEGV`	Core	Invalid memory reference
12	`SIGUSR2`	Term	User-defined signal 2
13	`SIGPIPE`	Term	Broken pipe
14	`SIGALRM`	Term	Timer signal from `alarm()`
15	`SIGTERM`	Term	Termination signal
16	`SIGSTKFLT`	Term	Stack fault on coprocessor (unused)

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Linux Signals: Default Actions (2/2)

#	Signal	Default Action	Description
17	`SIGCHLD`	Ign	Child stopped or terminated
18	`SIGCONT`	Cont	Continue if stopped
19	`SIGSTOP`	Stop (Unblockable)	Stop process
20	`SIGTSTP`	Stop	Stop from terminal (`Ctrl` + `Z`)
21	`SIGTTIN`	Stop	Background read from tty
22	`SIGTTOU`	Stop	Background write to tty
23	`SIGURG`	Ign	Urgent condition on socket
24	`SIGXCPU`	Core	CPU time limit exceeded
25	`SIGXFSZ`	Core	File size limit exceeded
26	`SIGVTALRM`	Term	Virtual timer expired
27	`SIGPROF`	Term	Profiling timer expired
28	`SIGWINCH`	Ign	Window resize signal
29	`SIGIO`	Term	I/O now possible
30	`SIGPWR`	Term	Power failure
31	`SIGSYS`	Core	Bad system call
34–64	`SIGRTMIN` – `SIGRTMAX`	Term	Real-time signals (implementation-defined)

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Foreground & Background

Run more than one process from the same shell

Bibliography

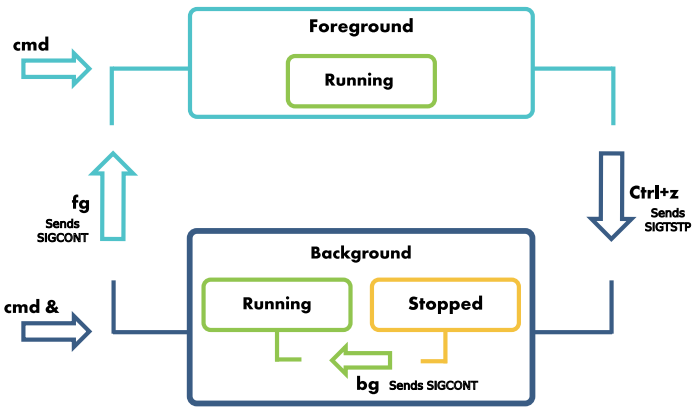
For this section

1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 2 - *Basic Commands and Directory Hierarchy*
     - Section 2.16 - *Listing and Manipulating Processes*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 4 - *Procese*
     - Section 4.3.3 - *Foreground și background*
     - Section 4.5.2 - *Semnale*

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Foreground and Background Process

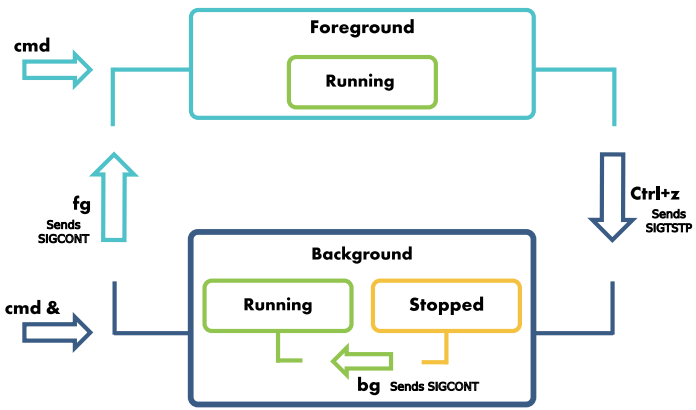
The terminal and the shell define this



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Foreground and Background Process

The terminal and the shell define this



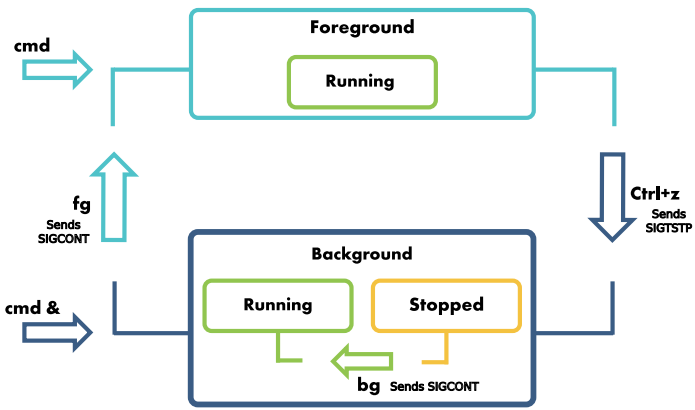
Foreground

- **only one** process
- reads the keyboard (`stdin`)
- the shell *waits* for it

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Foreground and Background Process

The terminal and the shell define this



Foreground

- **only one** process
- reads the keyboard (`stdin`)
- the shell *waits* for it

Background

- **any number** of processes, called *jobs*
- *running* or *stopped*
- the shell does not wait, it shows the prompt

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Start a Job

Add `&` at the end: the shell starts the command and shows the prompt **right away**

```
$ sleep 100 &
[1] 4242
```



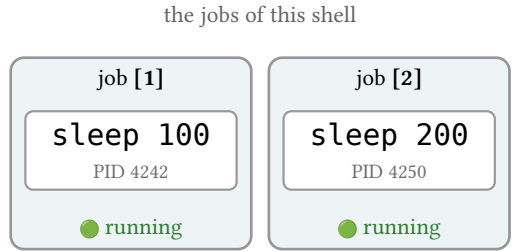
💡 `[1]` is the **job ID**, it exists only in this shell, `4242` is the **PID**

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Start a Job

Add `&` at the end: the shell starts the command and shows the prompt **right away**

```
$ sleep 100 &
[1] 4242
$ sleep 200 &
[2] 4250
```



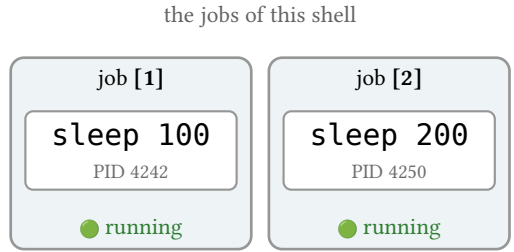
💡 a second job, `[2]`: the shell does not wait, both run at the same time

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Start a Job

Add `&` at the end: the shell starts the command and shows the prompt **right away**

```
$ sleep 100 &
[1] 4242
$ sleep 200 &
[2] 4250
$ jobs
[1]-  Running     sleep 100 &
[2]+  Running     sleep 200 &
```



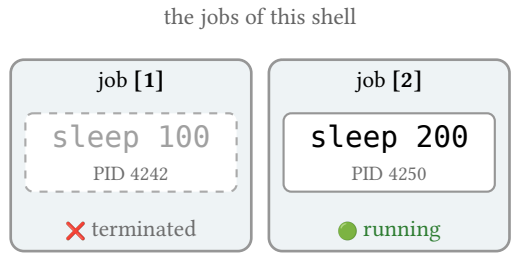
💡 `jobs` lists the jobs of **this** shell: `+` the current job (the last one), `-` the previous one

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Start a Job

Add `&` at the end: the shell starts the command and shows the prompt **right away**

```
$ sleep 100 &
[1] 4242
$ sleep 200 &
[2] 4250
$ jobs
[1]-  Running     sleep 100 &
[2]+  Running     sleep 200 &
$ kill %1
[1]-  Terminated  sleep 100
```



💡 `%1` means *job 1*: it works with `kill`, `fg` and `bg`, job `[2]` keeps running

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Make a Job

Move a foreground process to the background

- run the command
  - ⚠️ redirect its output, a job still writes to the same terminal

```
$ sleep 100 > /dev/null
```

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Make a Job

Move a foreground process to the background

- run the command
  - ⚠️ redirect its output, a job still writes to the same terminal
- press `Ctrl` + `Z` (sends `SIGTSTP`)
  - 🙄 the process might ignore it

```
$ sleep 100 > /dev/null
^Z
[1]+  Stopped     sleep 100 > /dev/null
```



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Make a Job

Move a foreground process to the background

- run the command
  - ⚠️ redirect its output, a job still writes to the same terminal
- press `Ctrl` + `Z` (sends `SIGTSTP`)
  - 🙄 the process might ignore it
- find the job `ID` (`jobs`)

```
$ sleep 100 > /dev/null
^Z
[1]+  Stopped     sleep 100 > /dev/null
$ jobs
[1]+  Stopped     sleep 100 > /dev/null
```



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Make a Job

Move a foreground process to the background

- run the command
  - ⚠️ redirect its output, a job still writes to the same terminal
- press `Ctrl` + `Z` (sends `SIGTSTP`)
  - 🙄 the process might ignore it
- find the job `ID` (`jobs`)
- run `bg %ID` (sends `SIGCONT`)

```
$ sleep 100 > /dev/null
^Z
[1]+  Stopped     sleep 100 > /dev/null
$ jobs
[1]+  Stopped     sleep 100 > /dev/null
$ bg %1
[1]+ sleep 100 > /dev/null &
```



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Make a Job

Move a foreground process to the background

- run the command
  - ⚠️ redirect its output, a job still writes to the same terminal
- press `Ctrl` + `Z` (sends `SIGTSTP`)
  - 🙄 the process might ignore it
- find the job `ID` (`jobs`)
- run `bg %ID` (sends `SIGCONT`)
- run `jobs` to verify if it still runs

```
$ sleep 100 > /dev/null
^Z
[1]+  Stopped     sleep 100 > /dev/null
$ jobs
[1]+  Stopped     sleep 100 > /dev/null
$ bg %1
[1]+ sleep 100 > /dev/null &
$ jobs
[1]+  Running     sleep 100 > /dev/null &
```



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Make a Job

Move a foreground process to the background

- run the command
  - ⚠️ redirect its output, a job still writes to the same terminal
- press `Ctrl` + `Z` (sends `SIGTSTP`)
  - 🙄 the process might ignore it
- find the job `ID` (`jobs`)
- run `bg %ID` (sends `SIGCONT`)
- run `jobs` to verify if it still runs

```
$ sleep 100 > /dev/null
^Z
[1]+  Stopped     sleep 100 > /dev/null
$ jobs
[1]+  Stopped     sleep 100 > /dev/null
$ bg %1
[1]+ sleep 100 > /dev/null &
$ jobs
[1]+  Running     sleep 100 > /dev/null &
$ fg %1
sleep 100 > /dev/null
```

Bring it back in the foreground

- run `fg %ID`

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Sessions and Terminals

How the processes of a terminal are grouped

Bibliography

For this section

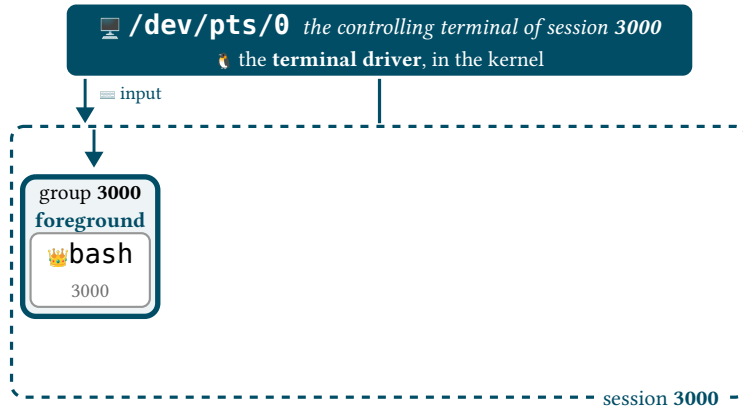
1. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 4 - *Procese*
     - Section 4.3.3 - *Foreground și background*
     - Section 4.5.2 - *Semnale*

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Sessions and the Terminal

How the kernel groups the processes of a terminal

```
$ echo $$
3000
```



👑 the shell starts a new **session**: a session has **one** controlling terminal, `/dev/pts/0`

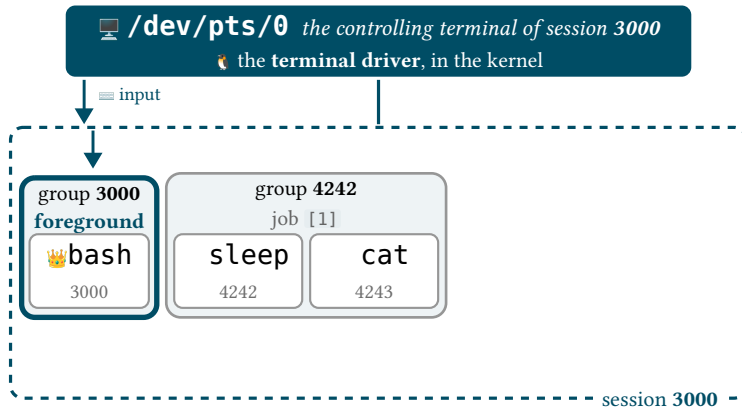
⌨️ the input of the terminal goes to the **foreground** group, now `bash` (`$$` is its PID)

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Sessions and the Terminal

How the kernel groups the processes of a terminal

```
$ echo $$
3000
$ sleep 100 | cat &
[1] 4243
```



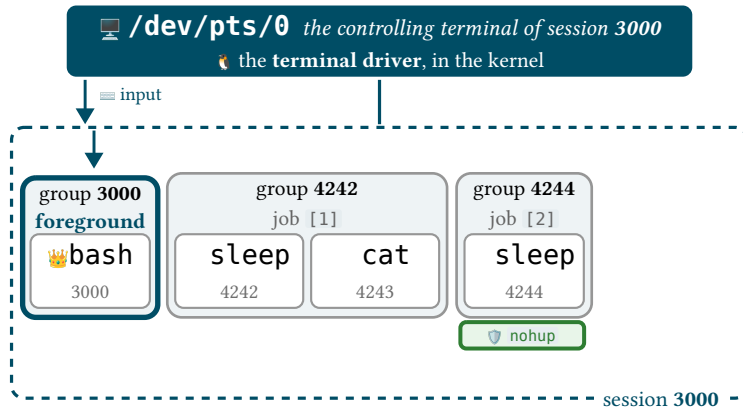
- a **process group**: one job, `sleep 100 | cat` is **one** group of two processes
- a signal (Ctrl + C) reaches the whole group

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Sessions and the Terminal

How the kernel groups the processes of a terminal

```
$ echo $$
3000
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
```



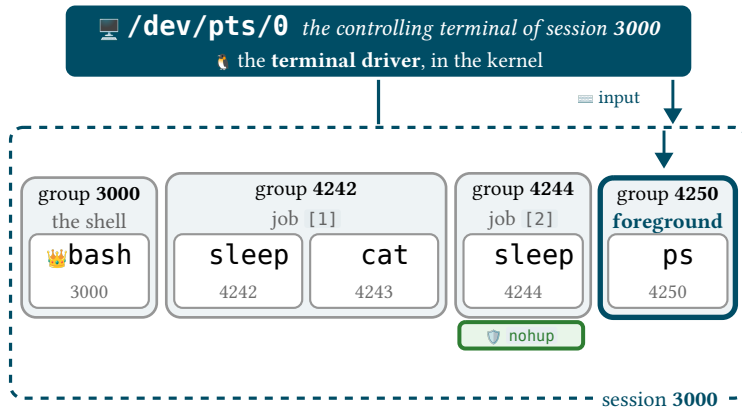
`nohup` makes the program **ignore** `SIGHUP`: a shield against it

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Sessions and the Terminal

How the kernel groups the processes of a terminal

```
$ echo $$
3000
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
$ ps -o pid,pgid,sid,stat,comm
    PID    PGID     SID STAT COMMAND
   3000    3000    3000 Ss   bash
   4242    4242    3000 S    sleep
   4243    4242    3000 S    cat
   4244    4244    3000 S    sleep
   4250    4250    3000 R+   ps
```



💡 a **session**: all the groups of one terminal, the same `SID`

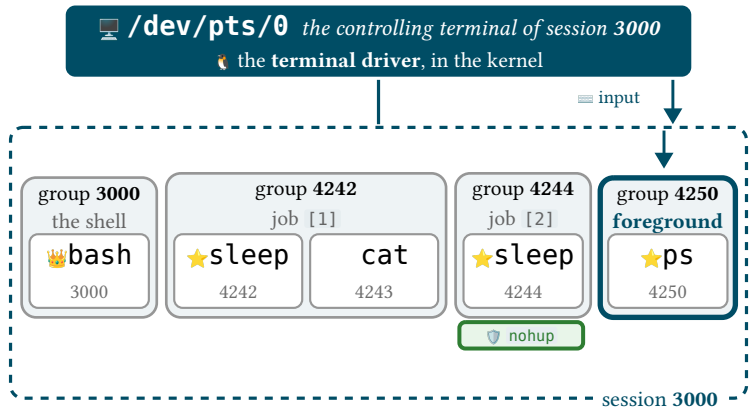
⌨️ only the **foreground** reads the input, `+` in `STAT`; a background job that reads gets `SIGTTIN` (stopped)

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Sessions and the Terminal

How the kernel groups the processes of a terminal

```
$ echo $$
3000
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
$ ps -o pid,pgid,sid,stat,comm
    PID    PGID     SID STAT COMMAND
   3000    3000    3000 Ss   bash
   4242    4242    3000 S    sleep
   4243    4242    3000 S    cat
   4244    4244    3000 S    sleep
   4250    4250    3000 R+   ps
```



session leader: its PID is the `SID` **group leader**: its PID is the `PGID`

the **group leader**: the first process of the group, its PID is the `PGID`

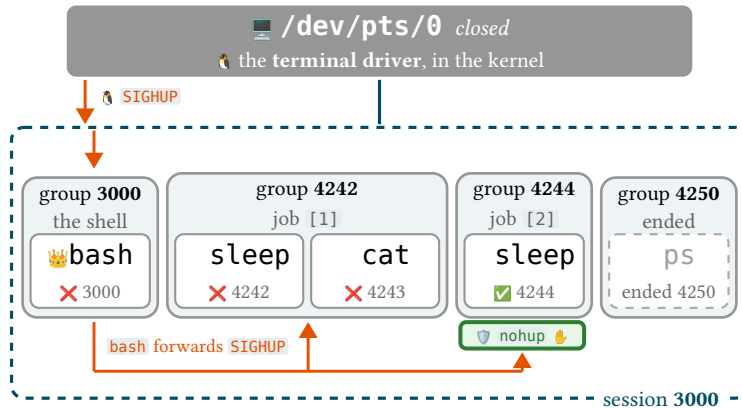
the **session leader**: the process that started the session, its PID is the `SID`, `s` in `STAT`

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Sessions and the Terminal

How the kernel groups the processes of a terminal

```
$ echo $$
3000
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
$ ps -o pid,pgid,sid,stat,comm
    PID    PGID     SID STAT COMMAND
   3000    3000    3000 Ss   bash
   4242    4242    3000 S    sleep
   4243    4242    3000 S    cat
   4244    4244    3000 S    sleep
   4250    4250    3000 R+   ps
```



when the terminal closes, the **terminal driver** sends `SIGHUP` to the session leader, `bash`

`bash` sends it to its jobs: job `[1]` ends, job `[2]` ignores it and keeps running

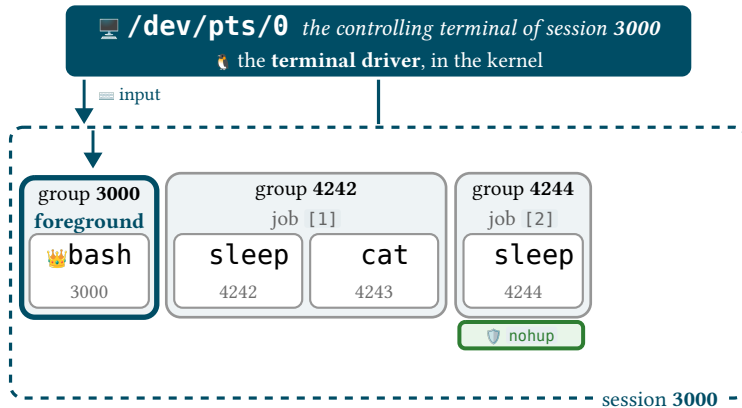
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When `bash` Is Killed

`SIGKILL` cannot be caught: the shell cannot forward `SIGHUP` any more

🖥️ *terminal `/dev/pts/0` (`bash`, PID 3000)*

```
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
```



💡 two jobs in the session of `bash`, as before

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When `bash` Is Killed

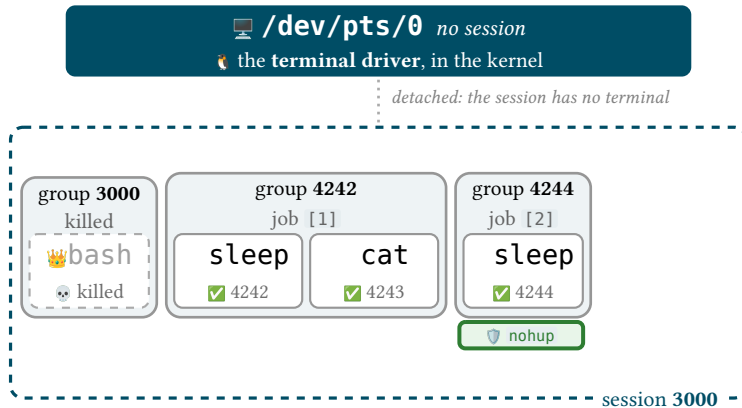
`SIGKILL` cannot be caught: the shell cannot forward `SIGHUP` any more

🖥️ *terminal `/dev/pts/0` (`bash`, PID 3000)*

```
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
```

🖥️ *another terminal, `/dev/pts/1`*

```
$ kill -KILL 3000
```



💀 `SIGKILL` cannot be caught: `bash` ends at once, it cannot forward anything

🐧 the session leader ended: the kernel detaches the terminal from the session

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When `bash` Is Killed

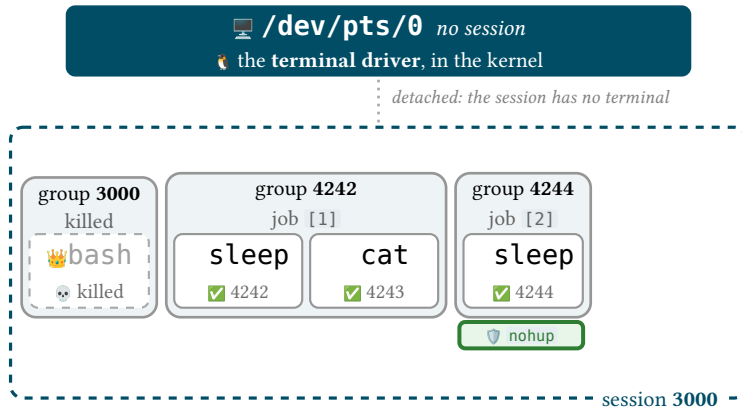
`SIGKILL` cannot be caught: the shell cannot forward `SIGHUP` any more

🖥️ *terminal `/dev/pts/0` (`bash`, PID 3000)*

```
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
```

🖥️ *another terminal, `/dev/pts/1`*

```
$ kill -KILL 3000
$ ps -o pid,ppid,sid,tty,comm -s 3000
    PID    PPID     SID TT       COMMAND
   4242       1    3000 ?        sleep
   4243       1    3000 ?        cat
   4244       1    3000 ?        sleep
```



💡 the jobs keep running: no controlling terminal (`TT` is `?`), their new parent is PID 1

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When `bash` Is Killed

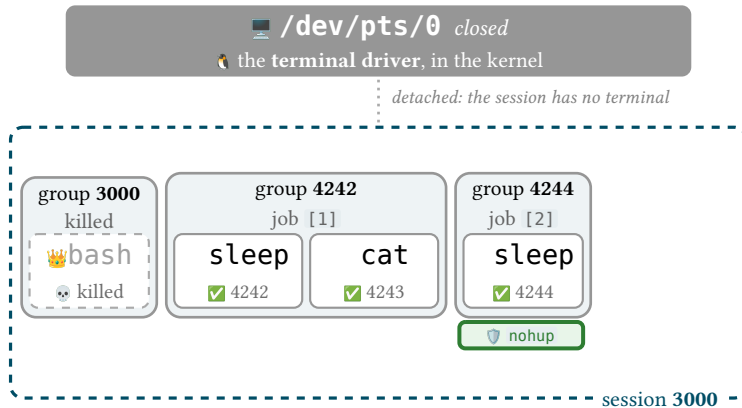
`SIGKILL` cannot be caught: the shell cannot forward `SIGHUP` any more

🖥️ *terminal `/dev/pts/0` (`bash`, PID 3000), closed*

```
$ sleep 100 | cat &
[1] 4243
$ nohup sleep 200 > /dev/null &
[2] 4244
nohup: ignoring input
```

🖥️ *another terminal, `/dev/pts/1`*

```
$ kill -KILL 3000
$ ps -o pid,ppid,sid,tty,comm -s 3000
    PID    PPID     SID TT       COMMAND
   4242       1    3000 ?        sleep
   4243       1    3000 ?        cat
   4244       1    3000 ?        sleep
```



🔌 the terminal closes: no `SIGHUP` reaches the jobs, they keep running in the background

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Keys and Signals

The terminal turns some keys into signals for the **foreground** process

Keys	Sends	Default action
`Ctrl` + `C`	`SIGINT`	*terminate* (interrupt) the process
`Ctrl` + `Z`	`SIGTSTP`	*stop* the process, it becomes a stopped job
`Ctrl` + `\`	`SIGQUIT`	*terminate* the process and write a core dump
`Ctrl` + `D`	no signal	*end of file*: `stdin` has nothing more to read

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Keys and Signals

The terminal turns some keys into signals for the **foreground** process

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`Ctrl` + `\`	`SIGQUIT`	*terminate* the process and write a core dump
`Ctrl` + `D`	no signal	*end of file*: `stdin` has nothing more to read

💡 a **job** that tries to read the keyboard receives `SIGTTIN` and is *stopped*, only the foreground process can read it

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Keys and Signals

The terminal turns some keys into signals for the **foreground** process

Keys	Sends	Default action
`Ctrl` + `C`	`SIGINT`	*terminate* (interrupt) the process
`Ctrl` + `Z`	`SIGTSTP`	*stop* the process, it becomes a stopped job
`Ctrl` + `\`	`SIGQUIT`	*terminate* the process and write a core dump
`Ctrl` + `D`	no signal	*end of file*: `stdin` has nothing more to read

💡 a **job** that tries to read the keyboard receives `SIGTTIN` and is *stopped*, only the foreground process can read it

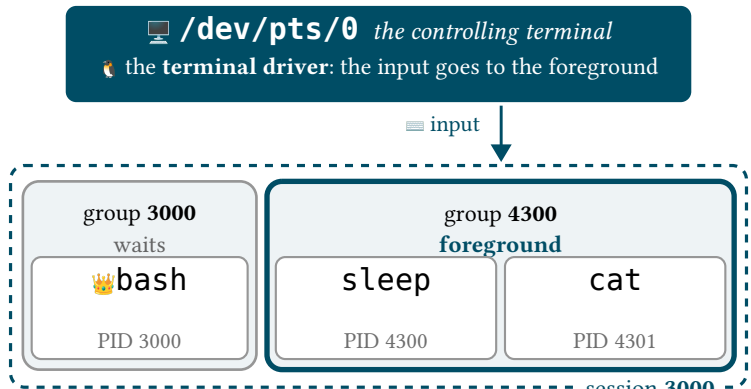
💡 `bg` and `fg` send `SIGCONT` to continue a stopped job

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Ctrl+C, Ctrl+\ and Ctrl+Z

The terminal driver turns three keys into signals for the **foreground** group

```
$ sleep 100 | cat
```



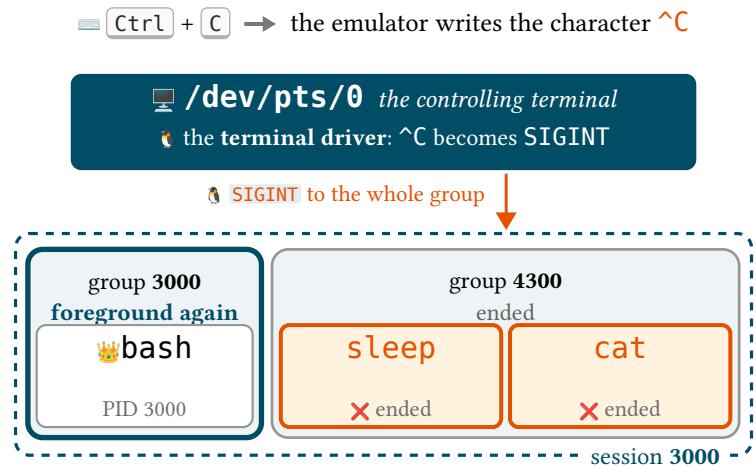
⌨️ `sleep 100 | cat` is the **foreground** group: it gets the input, `bash` waits for it

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Ctrl+C, Ctrl+\ and Ctrl+Z

The terminal driver turns three keys into signals for the **foreground** group

```
$ sleep 100 | cat
^C
```



the terminal driver sends `SIGINT` to **every** process of the foreground group

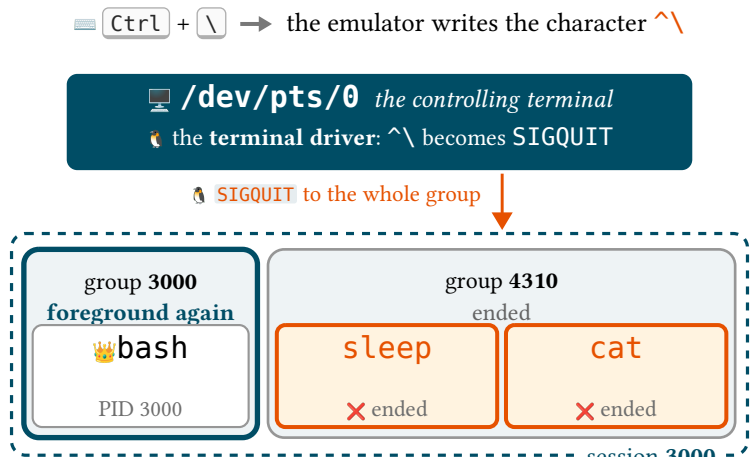
❌ `sleep` and `cat` end, `bash` gets the terminal back and shows the prompt

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Ctrl+C, Ctrl+\ and Ctrl+Z

The terminal driver turns three keys into signals for the **foreground** group

```
$ sleep 100 | cat
^C
$ sleep 200 | cat
^\Quit
```



🐧 a new job, `Ctrl` + `\` becomes `SIGQUIT`: the whole group ends, like `Ctrl` + `C`

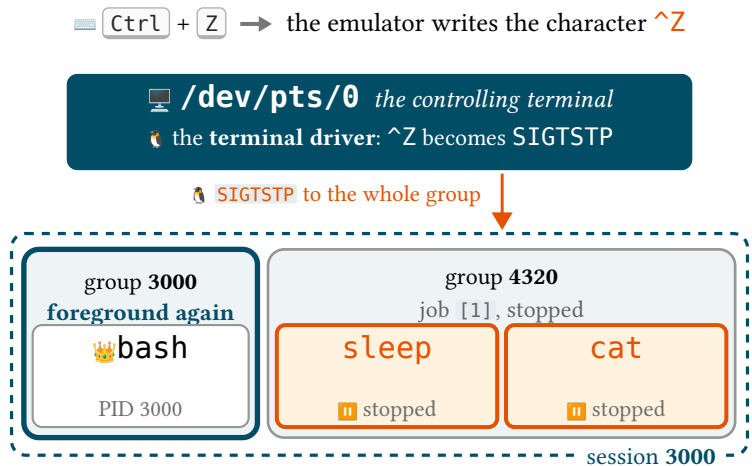
💡 it also writes a **core dump** (a copy of the memory, for debugging), if `ulimit -c` allows it

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Ctrl+C, Ctrl+\ and Ctrl+Z

The terminal driver turns three keys into signals for the **foreground** group

```
$ sleep 100 | cat
^C
$ sleep 200 | cat
^\Quit
$ sleep 300 | cat
^Z
[1]+  Stopped                 sleep 300 | cat
```



a new job, Ctrl + Z becomes `SIGTSTP`: the whole group is **stopped**, not ended

`bash` gets the terminal back, the group becomes job `[1]`, `fg` continues it

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Environment Variables

Settings that every process inherits

Bibliography

For this section

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   - Chapter 2 - *Basic Commands and Directory Hierarchy*
     - Section 2.8 - *Environment and Shell Variables*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 7 - *Interfața în linia de comandă*
     - Section 7.2.3 - *Variabile shell*

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Environment Variables

`NAME=value` texts that are part of every process

- every process has a list of `NAME=value` texts, its *environment*

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin
LANG	en_US.UTF-8
PWD	/home/alice/Movies

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Environment Variables

`NAME=value` texts that are part of every process

- every process has a list of `NAME=value` texts, its *environment*
- a process **receives** it from its parent: `fork` copies it, `exec` keeps it

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin
LANG	en_US.UTF-8
PWD	/home/alice/Movies

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Environment Variables

`NAME=value` texts that are part of every process

- every process has a list of `NAME=value` texts, its *environment*
- a process **receives** it from its parent: `fork` copies it, `exec` keeps it
- programs read it to configure themselves: `getenv("HOME")` in C, `$HOME` in `bash`

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin
LANG	en_US.UTF-8
PWD	/home/alice/Movies

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Environment Variables

`NAME=value` texts that are part of every process

- every process has a list of `NAME=value` texts, its *environment*
- a process **receives** it from its parent: `fork` copies it, `exec` keeps it
- programs read it to configure themselves: `getenv("HOME")` in C, `$HOME` in `bash`
- the names are written in **UPPER CASE**, by convention

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin
LANG	en_US.UTF-8
PWD	/home/alice/Movies

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Environment Variables

`NAME=value` texts that are part of every process

- every process has a list of `NAME=value` texts, its *environment*
- a process **receives** it from its parent: `fork` copies it, `exec` keeps it
- programs read it to configure themselves: `getenv("HOME")` in C, `$HOME` in `bash`
- the names are written in **UPPER CASE**, by convention

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin
LANG	en_US.UTF-8
PWD	/home/alice/Movies

```
$ printenv HOME    # one variable
/home/alice
$ echo "I am $USER" # the shell expands $USER
I am alice
$ env              # all the variables
```

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Inherited by the Children

Every process starts with a **copy** of the environment of its parent

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin

variables of `sort`

💡 `bash` has its own environment, like every process

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Inherited by the Children

Every process starts with a **copy** of the environment of its parent

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin

fork + exec



variables of `sort`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin

💡 the shell starts `sort`: the child gets a **copy** of all the environment variables

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Inherited by the Children

Every process starts with a **copy** of the environment of its parent

variables of `bash`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	/usr/local/bin:/usr/bin:/bin

fork + exec →

variables of `sort`	
HOME	/home/alice
USER	alice
SHELL	/bin/bash
PATH	**/tmp**

The child process can change **its own copy** using `setenv("PATH", "/tmp")` or `PATH=/tmp`, the parent does not see the change

💡 a process can only pass variables **down** to its children, never **up** to its parent

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export

Shell variables and environment variables: only the **exported** ones are passed to the children

```
$ GREETING=hello    # a shell variable
```



your shell

variables of **bash**		
HOME	/home/alice	
GREETING	*hello*	*(shell only)*

💡 `NAME=value` makes a **shell variable**: it exists only inside this shell

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export

Shell variables and environment variables: only the **exported** ones are passed to the children

```
$ GREETING=hello    # a shell variable
$ echo $GREETING
hello
```

bash

your shell

variables of **bash**		
HOME	/home/alice	
GREETING	*hello*	*(shell only)*

💡 `$NAME` is replaced by its value, by the shell, before the command starts

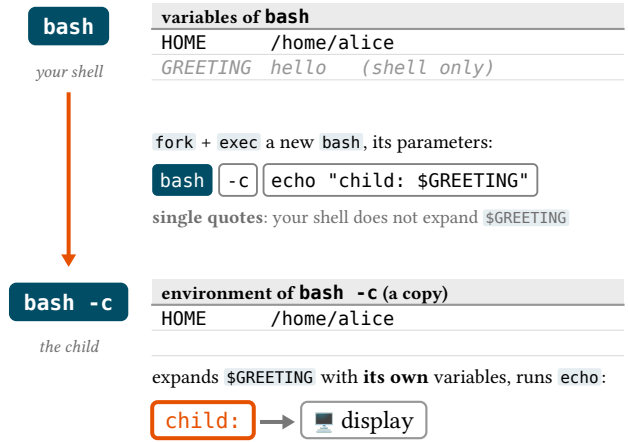
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export

Shell variables and environment variables: only the **exported** ones are passed to the children

```
$ GREETING=hello    # a shell variable
$ echo $GREETING
hello

# run a new bash instance that runs
#  echo "child: $GREETING" - mind the '...'
$ bash -c 'echo "child: $GREETING"'
child:
```



💡 the child (`bash -c`) does **not** receive it, it is not part of the environment

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`export`

Shell variables and environment variables: only the **exported** ones are passed to the children

```
$ GREETING=hello    # a shell variable
$ echo $GREETING
hello

# run a new bash instance that runs
#  echo "child: $GREETING" - mind the '...'
$ bash -c 'echo "child: $GREETING"'
child:
$ export GREETING   # an environment variable
```

bash

your shell

variables of **bash**	
HOME	/home/alice
GREETING	**hello**

💡 `export` marks it as an **environment variable**

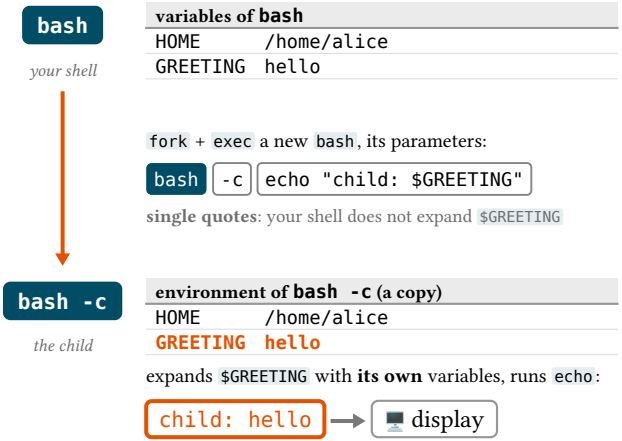
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`export`

Shell variables and environment variables: only the **exported** ones are passed to the children

```
$ GREETING=hello     # a shell variable
$ echo $GREETING
hello

# run a new bash instance that runs
#  echo "child: $GREETING" - mind the '...'
$ bash -c 'echo "child: $GREETING"'
child:
$ export GREETING    # an environment variable
$ bash -c 'echo "child: $GREETING"'
child: hello
```



- every child started from now on receives a copy
- `export NAME=value` does both at once, `unset NAME` deletes it

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Defined by the Shell

Variables that are already there when the prompt shows up

Variable	Value	Defined in (Fedora)
`HOME`, `USER`, `SHELL`	the home directory, the user, the shell	`/etc/passwd`, read by `login` when you log in
`PWD`, `OLDPWD`	the current and the previous directory	the shell itself, at every `cd` (used by `cd -`)
`PATH`	where commands are	`/etc/profile`, then `~/.bashrc` adds `~/.local/bin` and `~/bin`
`LANG`	the language	`/etc/locale.conf`, read by `/etc/profile.d/lang.sh`
`PS1`	what the prompt looks like, a **shell variable**	`/etc/bashrc`, read again by every interactive shell

💡 the startup files are shell scripts: they run `export NAME=value` (or `NAME=value` for `PS1`)

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Special Parameters

Set by the shell, they can only be read

- `$?` - the exit code of the last command (`0` means *success*)
- `$$` - the PID of the shell
- `$!` - the PID of the last job started with `&`

```
$ ls /nope
ls: cannot access '/nope': No such file or directory
```

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Special Parameters

Set by the shell, they can only be read

- `$?` - the exit code of the last command (`0` means *success*)
- `$$` - the PID of the shell
- `$!` - the PID of the last job started with `&`

```
$ ls /nope
ls: cannot access '/nope': No such file or directory
$ echo $?
2
```

💡 `ls` failed: the exit code is not `0`

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Special Parameters

Set by the shell, they can only be read

- `$?` - the exit code of the last command (`0` means *success*)
- `$$` - the PID of the shell
- `$!` - the PID of the last job started with `&`

```
$ ls /nope
ls: cannot access '/nope': No such file or directory
$ echo $?
2
$ echo $?    # exit code of echo
0
```

💡 `$?` changes after **every** command

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Special Parameters

Set by the shell, they can only be read

- `$?` - the exit code of the last command (`0` means *success*)
- `$$` - the PID of the shell
- `$!` - the PID of the last job started with `&`

```
$ ls /nope
ls: cannot access '/nope': No such file or directory
$ echo $?
2
$ echo $?    # exit code of echo
0
$ echo $$
4011
```

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Defined in the Command Line

Syntax	Defines
`NAME=value`	a **shell variable**, only for this shell
`export NAME=value`	an **environment variable**, for this shell and every command started after it
`NAME=value command`	an environment variable **only for** `command`, the shell does not keep it
`unset NAME`	deletes the variable

```
$ date +%H:%M
11:47
```

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Defined in the Command Line

Syntax	Defines
`NAME=value`	a **shell variable**, only for this shell
`export NAME=value`	an **environment variable**, for this shell and every command started after it
`NAME=value command`	an environment variable **only for** `command`, the shell does not keep it
`unset NAME`	deletes the variable

```
$ date +%H:%M
11:47
$ TZ=Asia/Tokyo date +%H:%M   # only for this date
17:47
```

💡 `TZ` (time zone) is set only in the environment of this `date` process

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Defined in the Command Line

Syntax	Defines
`NAME=value`	a **shell variable**, only for this shell
`export NAME=value`	an **environment variable**, for this shell and every command started after it
`NAME=value command`	an environment variable **only for** `command`, the shell does not keep it
`unset NAME`	deletes the variable

```
$ date +%H:%M
11:47
$ TZ=Asia/Tokyo date +%H:%M   # only for this date
17:47
$ echo "[$TZ]"                # the shell does not have it
[]
```

💡 useful to change a setting just once, e.g.
`LANG=C sort names.txt`

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Defined in the Command Line

Syntax	Defines
`NAME=value`	a **shell variable**, only for this shell
`export NAME=value`	an **environment variable**, for this shell and every command started after it
`NAME=value command`	an environment variable **only for** `command`, the shell does not keep it
`unset NAME`	deletes the variable

```
$ date +%H:%M
11:47
$ TZ=Asia/Tokyo date +%H:%M   # only for this date
17:47
$ echo "[$TZ]"                # the shell does not have it
[]
$ GREETING = hello
bash: GREETING: command not found
```

⚠️ **no spaces** around `=`: the shell thinks `GREETING` is a command

💡 use quotes for spaces in the value: `NAME="two words"`

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The Command Line

How the shell reads what you type

Bibliography

For this section

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     - Section 2.10 - *Special Characters*
     - Section 2.14 - *Shell Input and Output*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 7 - *Interfața în linia de comandă*
     - Section 7.2.3 - *Variabile shell*
     - Section 7.2.4 - *Expandări*

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A Command Line

Assignments, the command, its parameters and the redirects

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```

`LANG=C`
assignments
(optional)

💡 `NAME=value` words **at the start**: variables only for this command

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A Command Line

Assignments, the command, its parameters and the redirects

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```

`LANG=C` — assignments (optional)

`sort` — the command

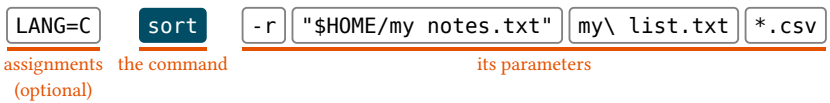
💡 the first word that is not an assignment is the **command**

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A Command Line

Assignments, the command, its parameters and the redirects

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```



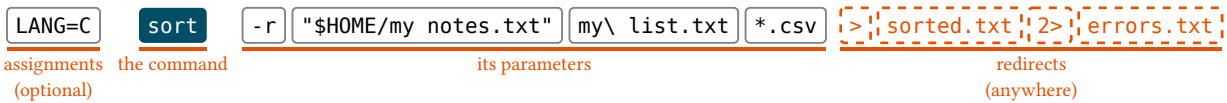
💡 everything else is a **parameter**: options, file names, texts

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A Command Line

Assignments, the command, its parameters and the redirects

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```



💡 a redirect (`>`, `2>`, `<`, ...) and its file name are for the shell, not for `sort`

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Splitting a Line

What the shell does, step by step, before the command starts

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```

1. split into words `LANG=C` `sort` `-r` `"$HOME/my notes.txt"` `my\ list.txt` `*.csv` `>` `sorted.txt` `2>` `errors.txt`

💡 split at the spaces, except inside quotes and after a `\`; `>` and `2>` are words of their own: 10 words

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Splitting a Line

What the shell does, step by step, before the command starts

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```

1. split into words

`LANG=C` | `sort` | `-r` | `"$HOME/my notes.txt"` | `my\ list.txt` | `*.csv` | `>` | `sorted.txt` | `2>` | `errors.txt`

2. set apart

`sort` | `-r` | `"$HOME/my notes.txt"` | `my\ list.txt` | `*.csv`

`LANG=C`

`> sorted.txt`

`2> errors.txt`

💡 the assignment (no spaces around `=`) and the redirects (`>` or `2>` + the next word) are set apart

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Splitting a Line

What the shell does, step by step, before the command starts

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```

1. split into words

`LANG=C` | `sort` | `-r` | `"$HOME/my notes.txt"` | `my\ list.txt` | `*.csv` | `>` | `sorted.txt` | `2>` | `errors.txt`

2. set apart

`sort` | `-r` | `"$HOME/my notes.txt"` | `my\ list.txt` | `*.csv`

`LANG=C`

`> sorted.txt`

`2> errors.txt`

3. replace `$NAME`, `~`

`sort` | `-r` | `"/home/alice/my notes.txt"` | `my\ list.txt` | `*.csv`

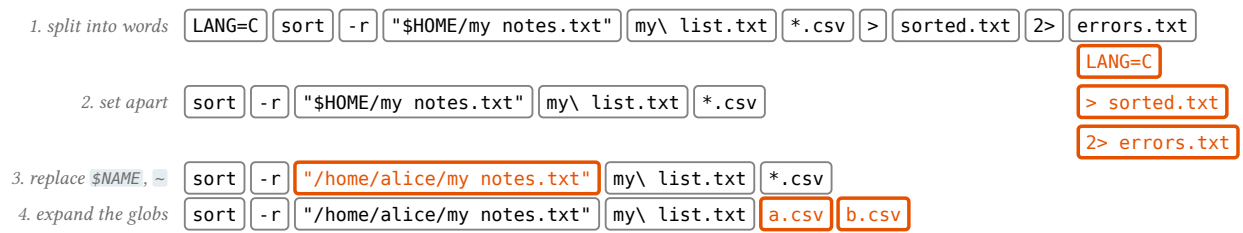
💡 `$HOME` is replaced inside `"..."` (not inside `'...'`), the result stays **one** word

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Splitting a Line

What the shell does, step by step, before the command starts

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```



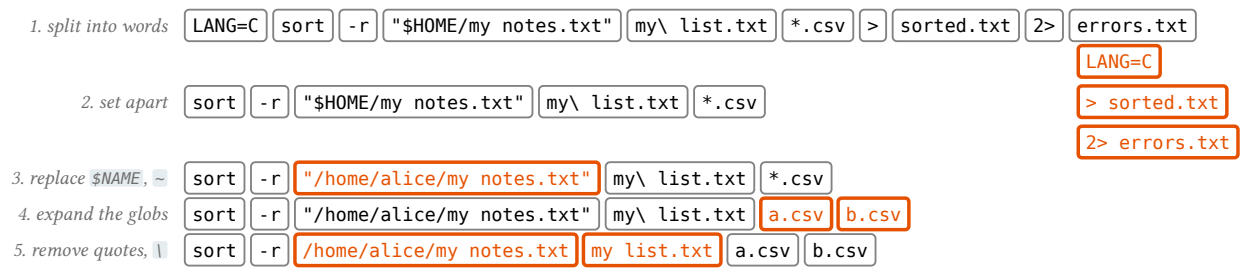
💡 `*.csv` is not quoted: the shell replaces it with the matching file names

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Splitting a Line

What the shell does, step by step, before the command starts

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```



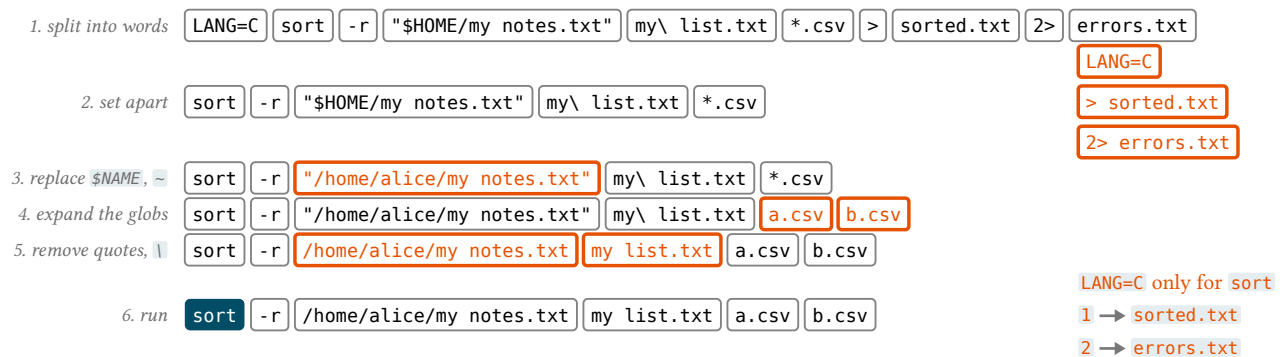
💡 the quotes and the `\` did their job: the shell removes them

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Splitting a Line

What the shell does, step by step, before the command starts

```
$ LANG=C sort -r "$HOME/my notes.txt" my\ list.txt *.csv > sorted.txt 2> errors.txt
```



💡 the shell sets up the redirects (it creates `sorted.txt`), then starts `sort` with 5 parameters and `LANG=C`

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Mixed Order

The parts can be mixed: only the assignments must be at the start

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



💡 `LANG=C` is at the start: an assignment, only for this command

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Mixed Order

The parts can be mixed: only the assignments must be at the start

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



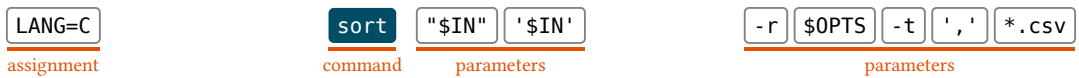
💡 the first word that is not an assignment or a redirect is the **command**

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Mixed Order

The parts can be mixed: only the assignments must be at the start

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



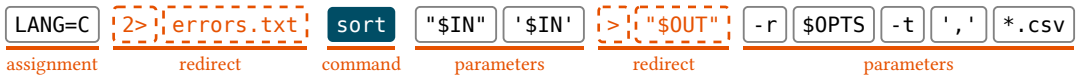
💡 the parameters do not have to be together, a redirect can be between them

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Mixed Order

The parts can be mixed: only the assignments must be at the start

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



💡 the redirects can be anywhere: even before the command, or between the parameters

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```

1. split into words `LANG=C` `2>` `errors.txt` `sort` `"$IN"` `'$IN'` `>` `"$OUT"` `-r` `$OPTS` `-t` `','` `*.csv`

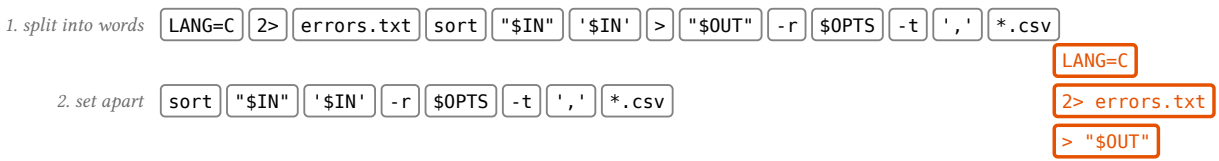
💡 the first line sets `IN`, `OUT` and `OPTS`; the second one has 13 words, a redirect is even before the command

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



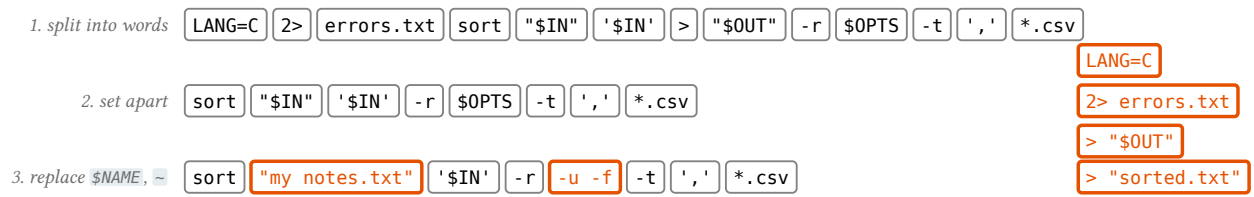
💡 the assignment (at the start) and the 2 redirects are set apart; `> "$OUT"` was in the **middle**

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



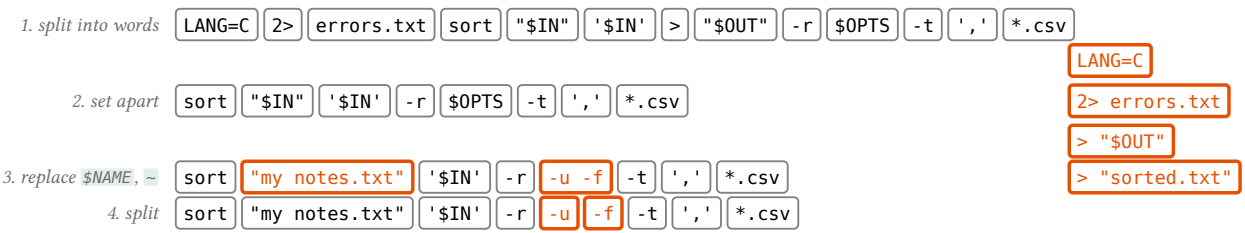
💡 `$IN` is replaced inside `"..."`, but **not** inside `'...'`; `$OPTS` and `$OUT` are replaced too

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



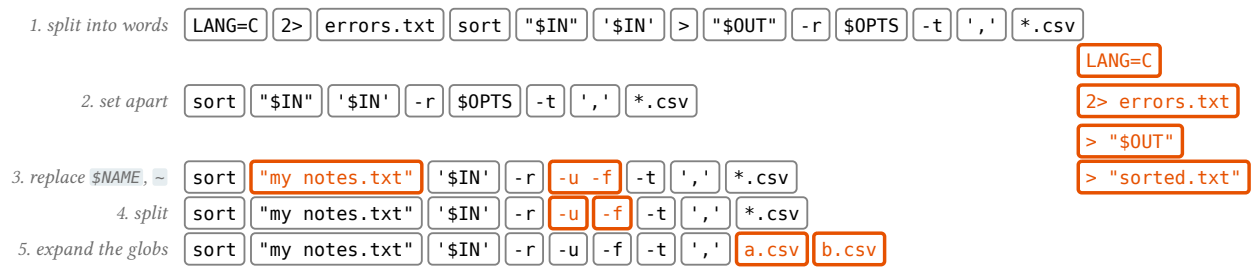
💡 `$OPTS` was not quoted: it becomes 2 words; `"$IN"` was quoted: `my notes.txt` stays one word

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



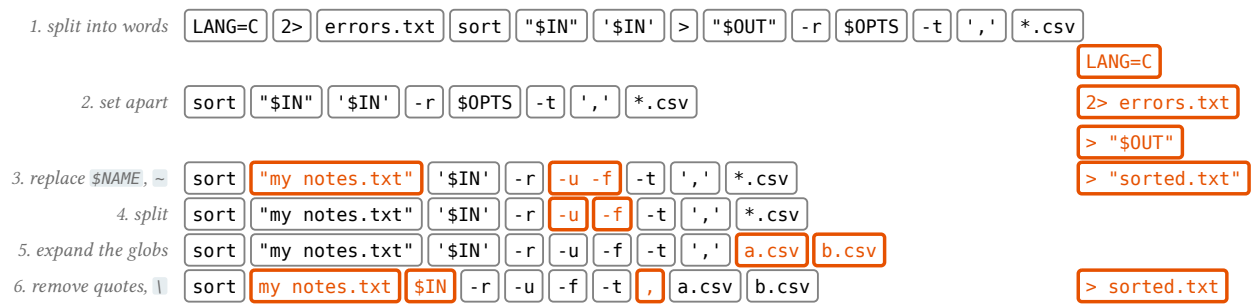
💡 `*.csv` is not quoted: the files of the current directory, `a.csv` and `b.csv`

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



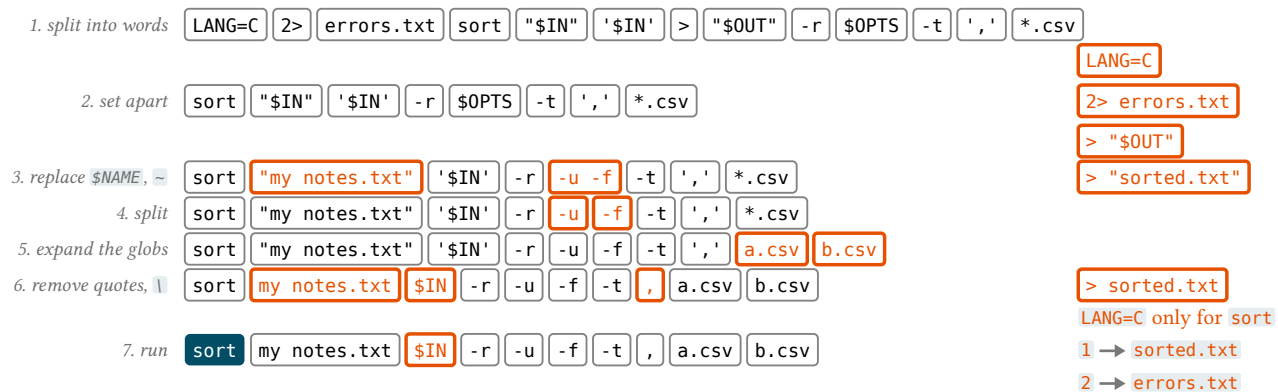
💡 the quotes did their job: the shell removes them, `'$IN'` becomes the text `$IN`

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Another Line

The assignment at the start, the redirects anywhere, double and single quotes, a glob

```
$ IN="my notes.txt" OUT=sorted.txt OPTS="-u -f"
$ LANG=C 2> errors.txt sort "$IN" '$IN' > "$OUT" -r $OPTS -t ',' *.csv
```



💡 `sort` gets a file named `$IN`: it does not exist, the error goes to `errors.txt`

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The Order of Redirects

The shell applies them **from left to right**

```
$ ls /nope > all.txt 2>&1
```

`1` → 🖥️ display
`2` → 🖥️ display

💡 `2>&1` means: `2` goes where `1` goes **now**

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The Order of Redirects

The shell applies them **from left to right**

```
$ ls /nope > all.txt 2>&1
```



💡 `2>&1` means: `2` goes where `1` goes **now**

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The Order of Redirects

The shell applies them **from left to right**

```
$ ls /nope > all.txt 2>&1
```



nothing on the display, the error is in `all.txt`

💡 `2>&1` means: `2` goes where `1` goes **now**

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The Order of Redirects

The shell applies them **from left to right**

```
$ ls /nope > all.txt 2>&1
```



nothing on the display, the error is in `all.txt`

```
$ ls /nope 2>&1 > all.txt
```



💡 `2>&1` is a **copy**, not a link: changing `1` later does not change `2`

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The Order of Redirects

The shell applies them **from left to right**

```
$ ls /nope > all.txt 2>&1
```



nothing on the display, the error is in `all.txt`

```
$ ls /nope 2>&1 > all.txt
```



💡 `2>&1` is a **copy**, not a link: changing `1` later does not change `2`

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The Order of Redirects

The shell applies them **from left to right**

```
$ ls /nope > all.txt 2>&1
```



nothing on the display, the error is in `all.txt`

```
$ ls /nope 2>&1 > all.txt
```



the error is on the **display**

💡 `2>&1` is a **copy**, not a link: changing `1` later does not change `2`

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Quoting, Escaping and Globbing

Special characters and file name patterns

Bibliography

For this section

1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 2 - *Basic Commands and Directory Hierarchy*
     - Section 2.4.2 - *Shell Globbing ("Wildcards")*
     - Section 2.10 - *Special Characters*
   - Chapter 11 - *Introduction to Shell Scripts*
     - Section 11.2 - *Quoting and Literals*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 7 - *Interfața în linia de comandă*
     - Sections 7.2.4 - 7.2.5 - *Expandări, Escaping*

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type → *the command and its parameters*

```
$ echo $NAME
```

→ `echo` | `Alice` | `Smith`

no quotes: replaced, then split at the spaces

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*	
`$ echo $NAME`	→	`echo` `Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo` `Alice   Smith`	double quotes: replaced, **one** word, spaces kept

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*	
`$ echo $NAME`	→	`echo` `Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo` `Alice   Smith`	double quotes: replaced, **one** word, spaces kept
`$ echo '$NAME'`	→	`echo` `$NAME`	single quotes: **nothing** is replaced

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*	
`$ echo $NAME`	→	`echo` `Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo` `Alice   Smith`	double quotes: replaced, **one** word, spaces kept
`$ echo '$NAME'`	→	`echo` `$NAME`	single quotes: **nothing** is replaced
`$ echo \$NAME`	→	`echo` `$NAME`	`\$`: the `$` is just a character

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*	
`$ echo $NAME`	→	`echo` `Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo` `Alice   Smith`	double quotes: replaced, **one** word, spaces kept
`$ echo '$NAME'`	→	`echo` `$NAME`	single quotes: **nothing** is replaced
`$ echo \$NAME`	→	`echo` `$NAME`	`\$`: the `$` is just a character
`$ echo my\ file.txt`	→	`echo` `my file.txt`	`\ `: the space does not split

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*	
`$ echo $NAME`	→	`echo` `Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo` `Alice   Smith`	double quotes: replaced, **one** word, spaces kept
`$ echo '$NAME'`	→	`echo` `$NAME`	single quotes: **nothing** is replaced
`$ echo \$NAME`	→	`echo` `$NAME`	`\$`: the `$` is just a character
`$ echo my\ file.txt`	→	`echo` `my file.txt`	`\ `: the space does not split
`$ echo "\$NAME is $NAME"`	→	`echo` `$NAME is Alice   Smith`	inside `"..."`, `\$` is a `$`, `$NAME` is replaced

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*		
`$ echo $NAME`	→	`echo`	`Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo`	`Alice   Smith`	double quotes: replaced, **one** word, spaces kept
`$ echo '$NAME'`	→	`echo`	`$NAME`	single quotes: **nothing** is replaced
`$ echo \$NAME`	→	`echo`	`$NAME`	`\$`: the `$` is just a character
`$ echo my\ file.txt`	→	`echo`	`my file.txt`	`\ `: the space does not split
`$ echo "\$NAME is $NAME"`	→	`echo`	`$NAME is Alice   Smith`	inside `"..."`, `\$` is a `$`, `$NAME` is replaced
`$ echo "it's" 'say "hi"'`	→	`echo`	`it's` `say "hi"`	each kind of quotes protects the other one

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Quoting and Escaping

With `NAME="Alice   Smith"`: the quotes and the `\` are gone, only the words are left

you type		*the command and its parameters*	
`$ echo $NAME`	→	`echo` `Alice` `Smith`	no quotes: replaced, then split at the spaces
`$ echo "$NAME"`	→	`echo` `Alice   Smith`	double quotes: replaced, **one** word, spaces kept
`$ echo '$NAME'`	→	`echo` `$NAME`	single quotes: **nothing** is replaced
`$ echo \$NAME`	→	`echo` `$NAME`	`\$`: the `$` is just a character
`$ echo my\ file.txt`	→	`echo` `my file.txt`	`\ `: the space does not split
`$ echo "\$NAME is $NAME"`	→	`echo` `$NAME is Alice   Smith`	inside `"..."`, `\$` is a `$`, `$NAME` is replaced
`$ echo "it's" 'say "hi"'`	→	`echo` `it's` `say "hi"`	each kind of quotes protects the other one
`$ echo 'a\b'`	→	`echo` `a\b`	inside `'...'`, `\` is just a character

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type

```
$ echo *.txt
```

→

the command and its parameters

```
echo notes.txt todo.txt
```

`*`: any characters, or none

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type → *the command and its parameters*

`$ echo *.txt` → `echo` `notes.txt` `todo.txt`

`$ echo photo?.png` → `echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`

`*`: any characters, or none

`?`: exactly one character

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type		*the command and its parameters*	
`$ echo *.txt`	→	`echo` `notes.txt` `todo.txt`	`*`: any characters, or none
`$ echo photo?.png`	→	`echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`	`?`: exactly one character
`$ echo photo[12].png`	→	`echo` `photo1.png` `photo2.png`	`[12]`: one of these characters

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type		*the command and its parameters*	
`$ echo *.txt`	→	`echo` `notes.txt` `todo.txt`	`*`: any characters, or none
`$ echo photo?.png`	→	`echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`	`?`: exactly one character
`$ echo photo[12].png`	→	`echo` `photo1.png` `photo2.png`	`[12]`: one of these characters
`$ echo photo[0-9].png`	→	`echo` `photo1.png` `photo2.png` `photo3.png`	`[0-9]`: one in the range

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type	*the command and its parameters*	
`$ echo *.txt`	`echo` `notes.txt` `todo.txt`	`*`: any characters, or none
`$ echo photo?.png`	`echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`	`?`: exactly one character
`$ echo photo[12].png`	`echo` `photo1.png` `photo2.png`	`[12]`: one of these characters
`$ echo photo[0-9].png`	`echo` `photo1.png` `photo2.png` `photo3.png`	`[0-9]`: one in the range
`$ echo photo[!0-9].png`	`echo` `photoA.png`	`[!0-9]`: one **not** in it

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type		*the command and its parameters*	
`$ echo *.txt`	→	`echo` `notes.txt` `todo.txt`	`*`: any characters, or none
`$ echo photo?.png`	→	`echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`	`?`: exactly one character
`$ echo photo[12].png`	→	`echo` `photo1.png` `photo2.png`	`[12]`: one of these characters
`$ echo photo[0-9].png`	→	`echo` `photo1.png` `photo2.png` `photo3.png`	`[0-9]`: one in the range
`$ echo photo[!0-9].png`	→	`echo` `photoA.png`	`[!0-9]`: one **not** in it
`$ echo docs/**/*.txt`	→	`echo` `docs/2026/exam.txt` `docs/plan.txt`	`**`: any directories (`globstar`)

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type	*the command and its parameters*	
`$ echo *.txt`	`echo` `notes.txt` `todo.txt`	`*`: any characters, or none
`$ echo photo?.png`	`echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`	`?`: exactly one character
`$ echo photo[12].png`	`echo` `photo1.png` `photo2.png`	`[12]`: one of these characters
`$ echo photo[0-9].png`	`echo` `photo1.png` `photo2.png` `photo3.png`	`[0-9]`: one in the range
`$ echo photo[!0-9].png`	`echo` `photoA.png`	`[!0-9]`: one **not** in it
`$ echo docs/**/*.txt`	`echo` `docs/2026/exam.txt` `docs/plan.txt`	`**`: any directories (`globstar`)
`$ echo {a,b,c}.pdf`	`echo` `a.pdf` `b.pdf` `c.pdf`	`{ }`: a word for each, file or not

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Globing

Patterns for file names, replaced by the shell with the matching files

```
$ ls
docs  notes.txt  photo1.png  photo2.png  photo3.png  photoA.png  report.md  todo.txt
```

you type		*the command and its parameters*	
`$ echo *.txt`	→	`echo` `notes.txt` `todo.txt`	`*`: any characters, or none
`$ echo photo?.png`	→	`echo` `photo1.png` `photo2.png` `photo3.png` `photoA.png`	`?`: exactly one character
`$ echo photo[12].png`	→	`echo` `photo1.png` `photo2.png`	`[12]`: one of these characters
`$ echo photo[0-9].png`	→	`echo` `photo1.png` `photo2.png` `photo3.png`	`[0-9]`: one in the range
`$ echo photo[!0-9].png`	→	`echo` `photoA.png`	`[!0-9]`: one **not** in it
`$ echo docs/**/*.txt`	→	`echo` `docs/2026/exam.txt` `docs/plan.txt`	`**`: any directories (`globstar`)
`$ echo {a,b,c}.pdf`	→	`echo` `a.pdf` `b.pdf` `c.pdf`	`{ }`: a word for each, file or not
`$ echo *.pdf`	→	`echo` `*.pdf`	no match: kept (`bash`), error (`zsh`)

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Running a Command

From a line of text to a process

Bibliography

For this section

1. **Brian Ward**, *How LINUX Works*, 3rd Edition, No Starch Press, 2021
   - Chapter 2 - *Basic Commands and Directory Hierarchy*
     - Section 2.9 - *The Command Path*
2. **Razvan Deaconescu, Razvan Rughinis, Mihai Carabas, Alexandru Radovici**, *Utilizarea Sistemelor de Operare*, Printech 2021, download
   - Chapter 7 - *Interfața în linia de comandă*
     - Section 7.1.3 - *Funcționarea shellului*
     - Section 7.1.3.1 - *Variabila de mediu PATH*
     - Section 7.1.4 - *Comenzi interne și comenzi externe*

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Built-in and External Commands

Not every command is a program

Built-in commands

- run **inside** the shell, no new process
- `cd`, `export`, `unset`, `jobs`, `fg`, `bg`, `type`, `exit`, ...

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Built-in and External Commands

Not every command is a program

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- run **inside** the shell, no new process
- `cd`, `export`, `unset`, `jobs`, `fg`, `bg`, `type`, `exit`, ...

External commands

- executable files, somewhere on the disk
- the shell needs their **path**
- `ls`, `sort`, `grep`, `bash`, ...

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Built-in and External Commands

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External commands

- executable files, somewhere on the disk
- the shell needs their **path**
- `ls`, `sort`, `grep`, `bash`, ...

```
$ type cd
cd is a shell builtin
$ type sort
sort is /usr/bin/sort
```

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Built-in and External Commands

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```
$ type cd
cd is a shell builtin
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sort is /usr/bin/sort
```

Why must `cd`, `unset` and `exit` be built-in?

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Built-in and External Commands

Not every command is a program

Built-in commands

- run **inside** the shell, no new process
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External commands

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```
$ type cd
cd is a shell builtin
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sort is /usr/bin/sort
```

Why must `cd`, `unset` and `exit` be built-in?

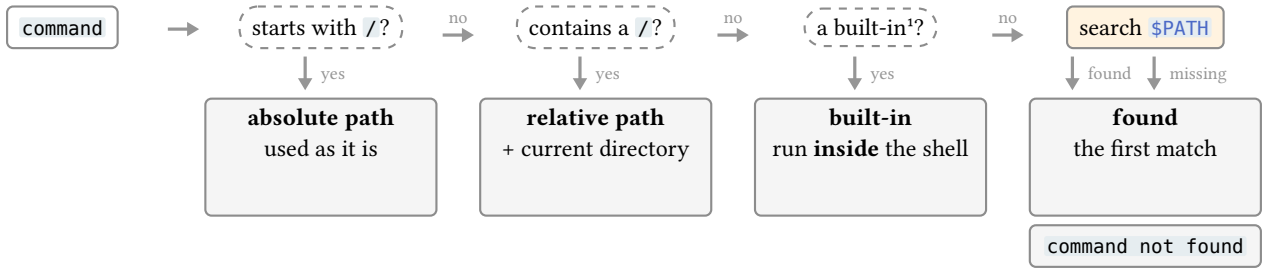
💡 they change **the shell itself**: `cd` its current directory, `unset` its variables, `exit` stops it

💡 a child process can only change **itself**, never its parent

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



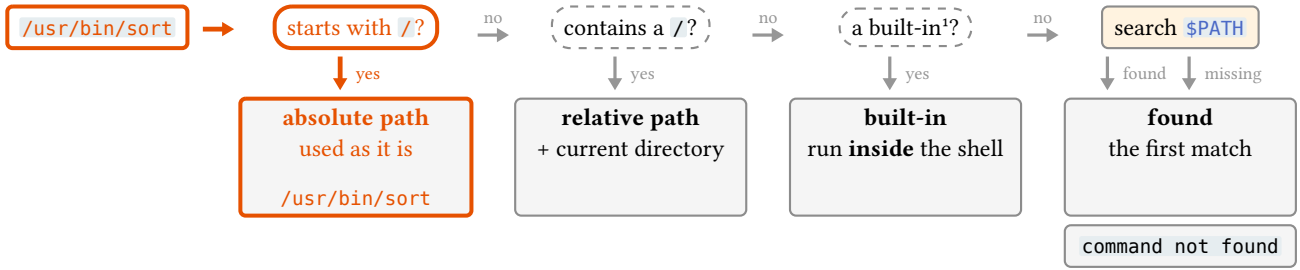
💡 the current directory is `/home/alice`

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ /usr/bin/sort fruits.txt
```

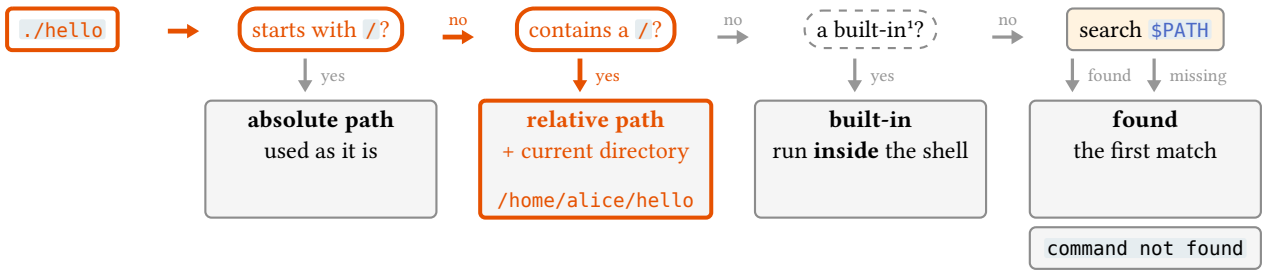
✅ an absolute path is used as it is, no search

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ ./hello
Hello!
```

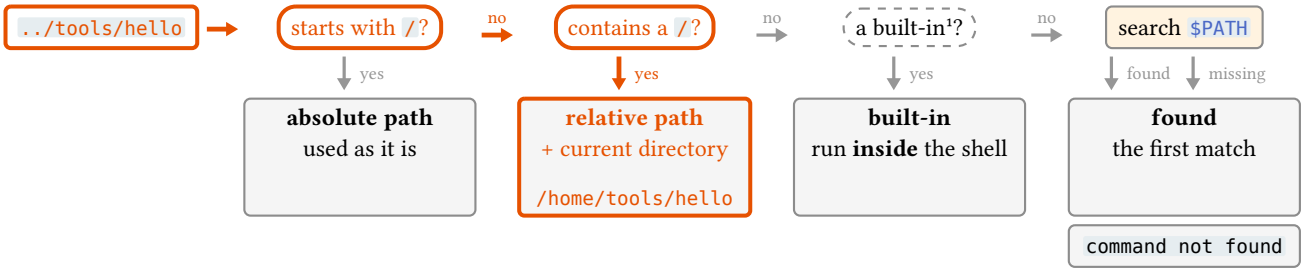
✅ `/home/alice` + `./hello` → `/home/alice/hello`

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ ../tools/hello
Hello from tools!
```

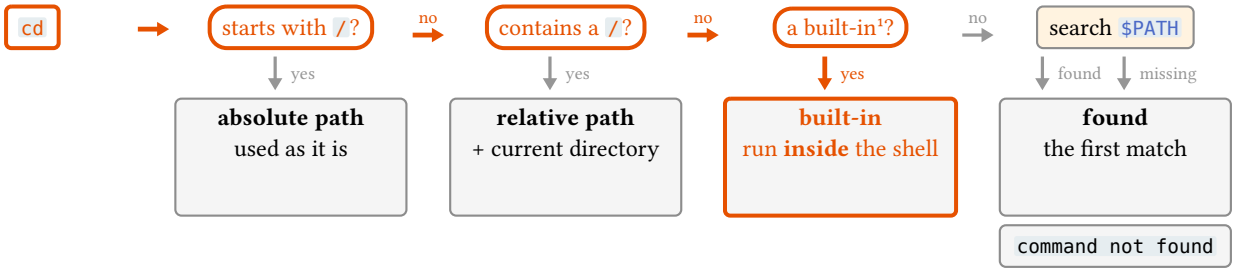
✅ `/home/alice` + `../tools/hello` → `/home/tools/hello`

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ cd /tmp
```

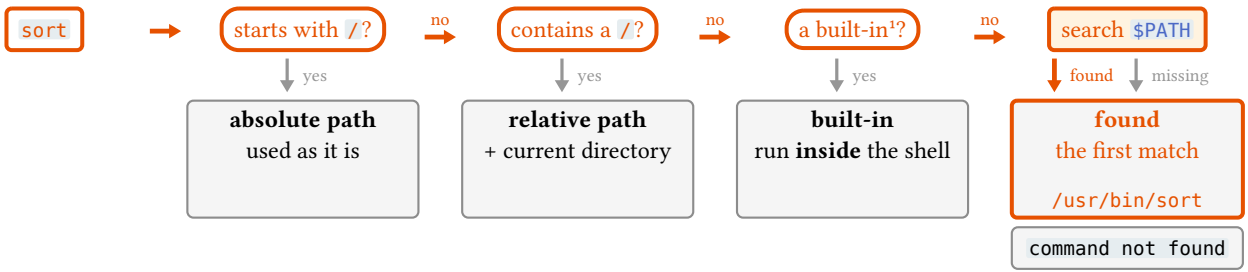
✅ `cd` is not a file, the shell does it itself

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ sort fruits.txt
```

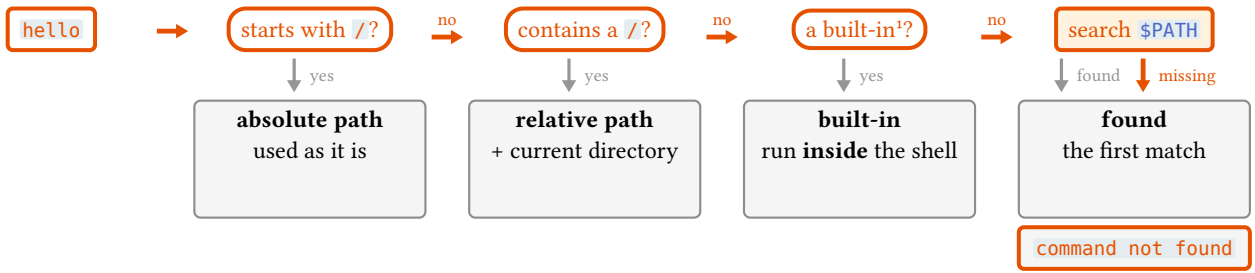
✅ the first match in `PATH`: `/usr/bin/sort`

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ hello
bash: hello: command not found
```

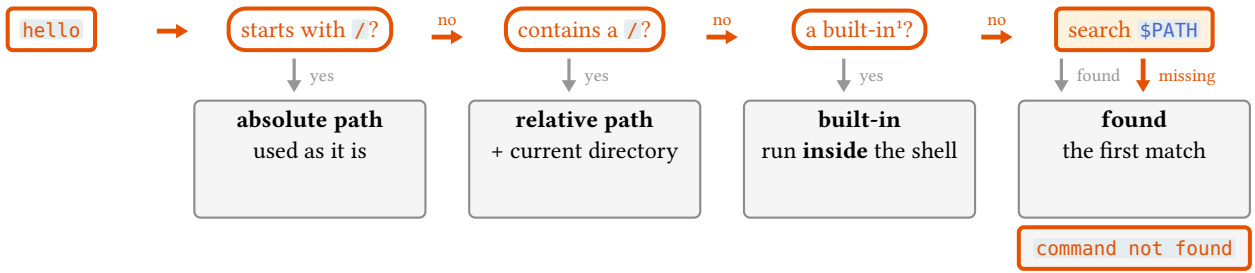
❌ no `hello` in `$PATH`: the shell does **not** look in the current directory

[1] it is actually a *function*, *alias* or *builtin*

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Which File?

External commands need a **path**: the shell turns the first word into a file to start



you type

```
$ hello
bash: hello: command not found
```

🐧 `exec` only starts a path: turning a name into a path is the job of the shell[2]

[1] it is actually a *function*, *alias* or *builtin*

[2] It is actually the job of `libc`, the C library, the shell uses the `execvp` function

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```

💡 a command name without a `/`: the shell has to find its executable file

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```

↓ *`sort` has no `/`: look it up in the environment variable `PATH`*

```
PATH=/home/alice/.local/bin:/usr/local/bin:/usr/bin:/bin
```

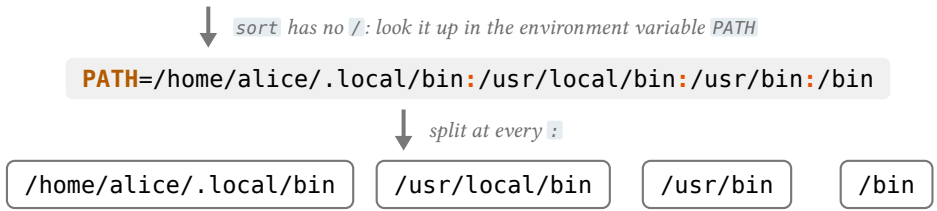
💡 `PATH` is an environment variable: a list of directories, separated by `:`

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```



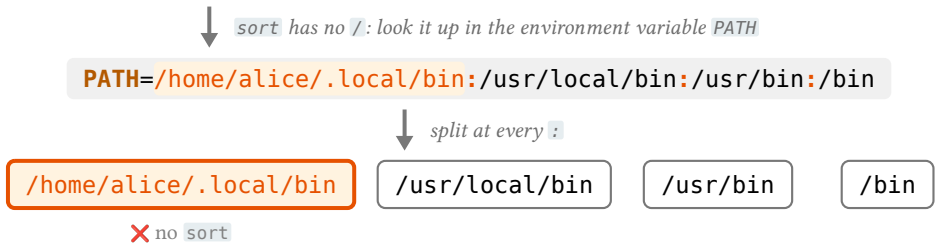
💡 the shell splits it into directories

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```



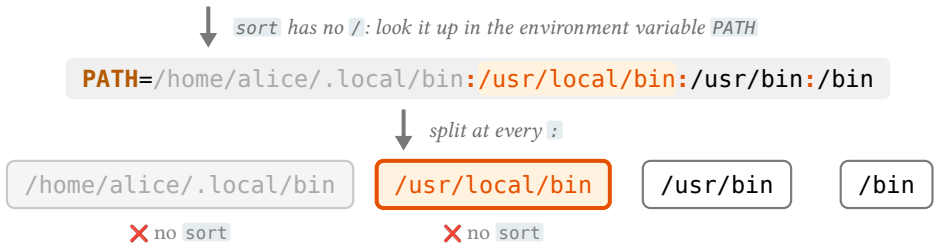
💡 it checks them **from left to right**: there is no `sort` in `/home/alice/.local/bin`

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```



💡 no `sort` in `/usr/local/bin` either

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```



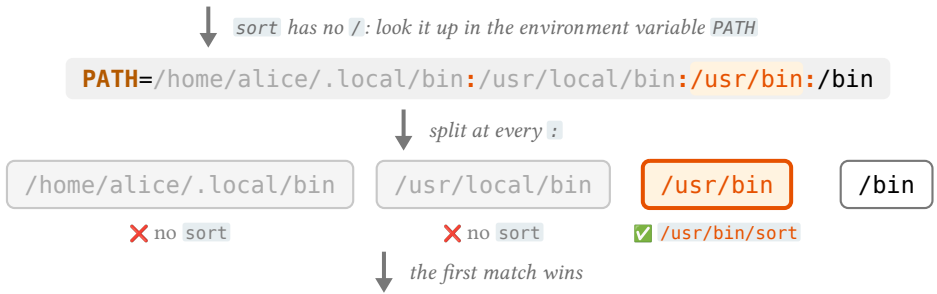
💡 `/usr/bin/sort` exists: the search stops, `/bin` is never checked

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`$PATH` Search

Where the shell looks for commands

```
$ sort fruits.txt
```




```
$ /usr/bin/sort fruits.txt
```

💡 the shell runs the file it found

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which

Do the `$PATH` search, without running the command

- prints the file the shell would run

```
$ which sort
/usr/bin/sort
```

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`which`

Do the `$PATH` search, without running the command

- prints the file the shell would run
- `-a` prints **every** match, in the order of `PATH`

```
$ which sort
/usr/bin/sort
```

```
$ which -a sort
/usr/bin/sort
/bin/sort
```

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`which`

Do the `$PATH` search, without running the command

- prints the file the shell would run
- `-a` prints **every** match, in the order of `PATH`
- nothing found: no output, exit status `1`

```
$ which sort
/usr/bin/sort
```

```
$ which -a sort
/usr/bin/sort
/bin/sort
```

```
$ which cd
$ echo $?
1
```

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`which`

Do the `$PATH` search, without running the command

- prints the file the shell would run
- `-a` prints **every** match, in the order of `PATH`
- nothing found: no output, exit status `1`

Why does `which` not find `cd`?

```
$ which sort
/usr/bin/sort
```

```
$ which -a sort
/usr/bin/sort
/bin/sort
```

```
$ which cd
$ echo $?
1
```

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`which`

Do the `$PATH` search, without running the command

- prints the file the shell would run
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- nothing found: no output, exit status `1`

```
$ which sort
/usr/bin/sort
```

```
$ which -a sort
/usr/bin/sort
/bin/sort
```

Why does `which` not find `cd`?

💡 `which` is an **external** command: it only searches `$PATH`, it does not know the built-ins of the shell

💡 `type cd` asks the shell itself:

`cd is a shell builtin`

```
$ which cd
$ echo $?
1
```

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We talked about

- Terminal & Shell
- Foreground & Background
- Sessions and Terminals
- Signals
- Environment Variables
- The Command Line
- Quoting, Escaping and Globbing
- Running a Command

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